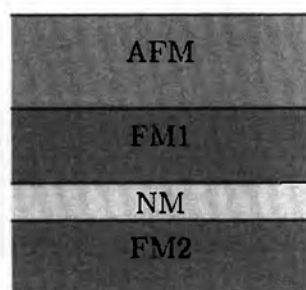


**Fig. 8.21** A hysteresis loop shifted by an exchange anisotropy. Without the exchange anisotropy the loop would be centred at zero magnetic field.



**Fig. 8.22** A spin valve. The antiferromagnet (AFM) 'clamps' the magnetic moment in the upper ferromagnetic layer (FM1) along a particular direction. The magnetic moment in the lower magnetic layer (FM2) is free to rotate and can be aligned with an applied magnetic field. The angle between the magnetic moments in the two ferromagnetic layers controls the resistance of the device. In zero applied magnetic field, the relative alignment of the layers is controlled by the thickness of the non-magnetic spacer layer (NM).

favourable for the ferromagnetic film to be magnetized one way (in the direction in which it was cooled) rather than the other. It is sometimes said to be exchange biased by the antiferromagnetic layer. The effect of the exchange biasing is to produce a unidirectional exchange field  $B_{ex}$  which can act in competition with the applied field  $B$  (the total energy of the magnetic layer is given by  $-M \cdot (B + B_{ex})$ ). If  $B$  and  $B_{ex}$  are in the same direction, the effect is simply to shift the hysteresis loop (see Fig. 8.21). If they are at right angles, a hard axis response is obtained. In both cases, there is a *unique* angle of the magnetization which minimizes the energy at each value of the magnetic field.

This technique can be used to add an 'exchange bias' field to magnetoresistive sensor heads in order to bias them into their linear regions. It is also used to constrain the direction of the magnetization in one soft ferromagnetic layer in a sandwich structure. The magnetization in the other layer can then be rotated by an applied field, allowing magnetoresistance and coupling energies to be measured in a carefully controlled way. This is also the basis of the spin-valve (see Fig. 8.22) which is a giant magnetoresistive sensor consisting of two magnetic layers with a non-magnetic spacer, with one of the magnetic layers adjacent to an antiferromagnetic layer. The resistance across the sandwich structure is then sensitive to the value of the applied magnetic field along a particular direction.

### 8.9.5 Colossal magnetoresistance

The transport properties of Mn oxides have recently generated enormous interest.  $\text{LaMnO}_3$  contains Mn in the  $\text{Mn}^{3+}$  state which is a Jahn–Teller ion.  $\text{LaMnO}_3$  shows A-type antiferromagnetic ordering (see Section 5.2.4). If a fraction  $x$  of the trivalent  $\text{La}^{3+}$  ions are replaced by divalent  $\text{Sr}^{2+}$ ,  $\text{Ca}^{2+}$  or  $\text{Ba}^{2+}$  ions, holes are introduced on the Mn sites. This results in a fraction  $1-x$  of the Mn ions remaining as  $\text{Mn}^{3+}$  ( $3d^4, t_{2g}^3 e_g^1$ ) and a fraction  $x$  becoming  $\text{Mn}^{4+}$  ( $3d^4, t_{2g}^3 e_g^0$ ). When  $x = 0.175$  the Jahn–Teller distortion vanishes and the system becomes ferromagnetic with a Curie temperature around room temperature. Above  $T_C$  the material is insulating and non-magnetic, but below  $T_C$  it is metallic and ferromagnetic. Particularly near  $T_C$  the material shows an extremely large magnetoresistive effect (see Fig. 8.23) which has been called colossal magnetoresistance (abbreviated to CMR), the term 'giant' having been already taken! The origin of the CMR is connected with the presence of a metal–insulator transition.

The origin of the behaviour described above is partly connected with the phenomenon of double exchange (see Section 4.2.5). In a  $\text{Mn}^{3+}$  ion, the  $t_{2g}$  electrons are tightly bound to the ion but the  $e_g$  electron is itinerant. Because of the double exchange interaction, the hopping of  $e_g$  electrons between Mn sites is only permitted if the two Mn core spins are aligned (in fact the hopping probability is proportional to  $|\cos(\theta/2)|$  where  $\theta$  is the angle between the two Mn core spins). The magnetic field aligns the core spins and therefore increases the conductivity, especially near  $T_C$ .

The situation is actually more complicated because the carriers interact with phonons because of the Jahn–Teller effect. The strong electron–phonon coupling in these systems implies that the carriers are actually polarons above  $T_C$ , i.e. electrons accompanied by a large lattice distortion. These polarons are

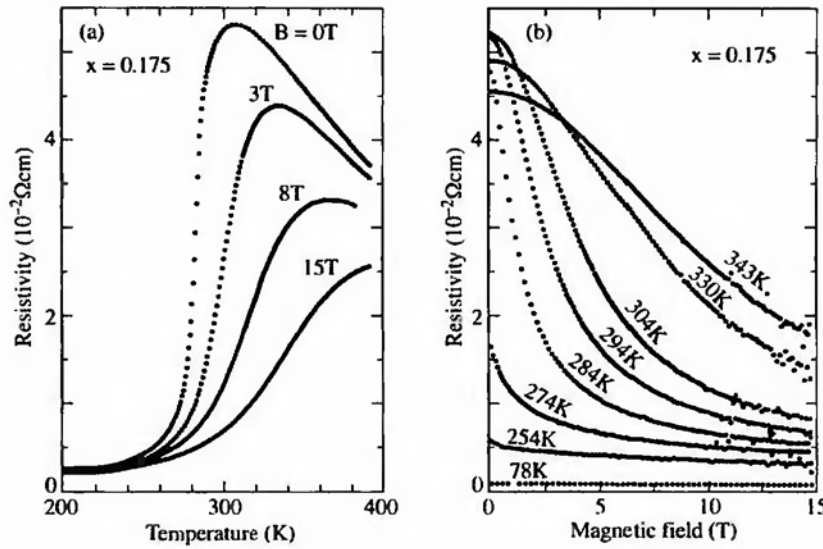


Fig. 8.23 Colossal magnetoresistance in  $\text{La}_{1-x}\text{Sr}_x\text{MnO}_3$  for  $x = 0.175$ . (a) Temperature dependence of the resistivity. (b) Isothermal magnetoresistance. After Tokura *et al.* 1994.

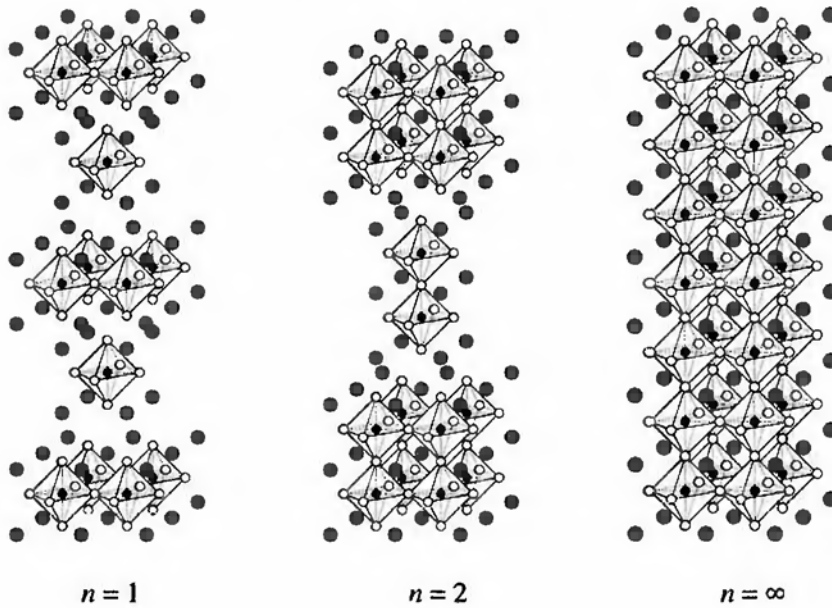


Fig. 8.24 The Ruddlesden-Popper phases of the manganites.

magnetic and are self-trapped in the lattice. The transition to the magnetic state can be regarded as an unbinding of the trapped polarons. There are other signatures of the electron-phonon couplings, including magnetic-field dependent structural transitions and charge ordering. The charge-ordered ground state can compete with ferromagnetism and is enhanced near commensurate values of the doping (e.g. at  $x = \frac{1}{2}$  where there is one hole for every two Mn sites). These effects are not fully understood and are under active current study.

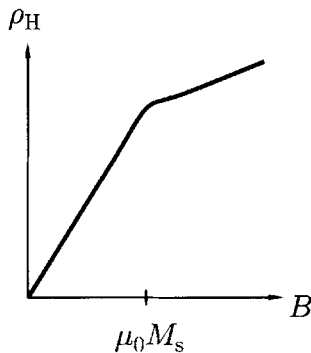
The manganese perovskites are just one type of oxide material which shows colossal magnetoresistance. Perovskites are a member of a large family of crystals known as Ruddlesden-Popper phases (see Fig. 8.24). These phases

<sup>8</sup>Perovskite blocks are essentially stacks of corner-sharing  $\text{MnO}_6$  octahedra.

have a general formula  $\text{X}_{n+1}\text{Mn}_n\text{O}_{3n+1}$  (where X is a lanthanide, strontium, or mixture) and can be thought of as stacks of perovskite blocks<sup>8</sup>  $n$  layers thick with each block separated by a rock-salt like  $(\text{Sr,Ln})_2\text{O}_2$  layer which tends to decouple the blocks electrically and magnetically. The perovskite compounds are realized when  $n = \infty$ , whereas the case  $n = 1$  is equivalent to the  $\text{K}_2\text{NiF}_4$  structure, adopted by the high- $T_c$  cuprates. Manganese compounds can be made which adopt a range of structures (e.g.  $n = 1, 2, 3$  and  $\infty$  have all been prepared at the time of writing). In the perovskites  $n = \infty$ , each  $\text{MnO}_6$  octahedron is surrounded by six others. In the  $n = 2$  phases, this 'coordination number' drops to five, and it is four for the  $n = 1$  phases. The reduction in the number of nearest-neighbours is expected to produce an anisotropic reduction in the width of the energy bands which are derived (largely) from the Mn 3d orbitals, and this modifies the electrical conductivity and magnetic behaviour of these materials. These materials can thus be carefully controlled by using different dopants, different doping levels, and adjusting the dimensionality.

### 8.9.6 Hall effect

<sup>9</sup>Usually this is simply called the Hall effect



**Fig. 8.25** The Hall resistivity in a ferromagnet.

A magnetic field applied normal to the current direction in a conducting material produces a transverse force on the conduction electrons in the film. This force on the conduction electrons gives rise to a transverse Hall voltage. This is known as the **ordinary Hall effect**<sup>9</sup> and is proportional to the applied magnetic field  $\mathbf{B}$ , because the Lorentz force on conduction electrons is  $\mathbf{F} = e(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ . In ferromagnets an additional effect occurs, known as the **extraordinary Hall effect** which depends on the magnetization. Note that it is *not* just a Lorentz force due to  $\mathbf{M}$  rather than  $\mathbf{B}$ , because this is already included in the ordinary Hall effect since  $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$ . Empirically the transverse resistivity  $\rho_H$  is given

$$\rho_H = R_0 B + \mu_0 R_e M, \quad (8.12)$$

where  $R_0$  is the ordinary Hall coefficient, and  $R_e$  is the extraordinary Hall coefficient (sometimes the spontaneous Hall coefficient). The ordinary Hall coefficient  $R_0$  tends to be fairly temperature independent, whereas  $R_e$  is usually very temperature dependent. The effect is not only seen in ferromagnets, but also in strong antiferromagnets or paramagnets. The effect only requires the presence of localized moments. (For example, it is seen in Tb in both its ferromagnetic and paramagnetic states.) If you measure the Hall resistivity as a function of field, a straight line graph results but there is a sudden change of gradient when the saturation field is reached (see Fig. 8.25). This is because at low magnetic field both  $B$  and  $M$  increase, so both ordinary and extraordinary Hall effects are seen and the gradient is due to  $R_0$  and  $R_e$ . Above the saturation field  $M$  can no longer show a further increase and the gradient results from  $R_0$  alone.

The mechanism for the extraordinary Hall effect is associated with the spin-orbit interaction between the conduction electrons and localized moments. The carriers have an orbital angular momentum about the scattering centre whose sign depends on whether they pass the centre on one side or the other. This orbital angular momentum  $\mathbf{L}$  couples to the spin angular momentum  $\mathbf{S}$  of the scattering centre because of the spin-orbit interaction which is proportional to  $\mathbf{L} \cdot \mathbf{S}$ . The electrons thus find it energetically favourable to pass to one side

rather than the other. This results in asymmetric scattering and hence to a transverse current which is responsible for the extraordinary Hall effect.

## 8.10 Organic and molecular magnets

Magnetic materials have been conventionally prepared from substances which are essentially the products of inorganic chemistry. However there has been recent interest in *organic* magnetic materials which are based upon organic chemistry. The tunability, resulting from the rich structure of carbon chemistry which allows many small adjustments to be made to each molecule, means that in principle materials can be tailor-made to exhibit desired properties. This is only 'in principle' however, since the structure–property relations can turn out to be remarkably intricate. The impetus to study organic materials in condensed matter physics stems particularly from the recent remarkable discoveries of three types of new material: first, conducting polymers which can have electrical conductivities as high as conventional metals such as copper and which have been used in the fabrication of polymer transistors and light-emitting-diode devices; second, superconducting charge transfer salts which can exhibit superconducting transition temperatures as high as 15 K; third, carbon 'bucky-balls'  $C_{60}$ , a new allotrope of carbon which when appropriately doped may exhibit superconductivity or even unusual magnetic properties.

Ferromagnets are rather rare even among the elements and are exclusively found in the d or f block. To achieve a magnetic moment on an organic material, it is natural to search for an organic free radical which has unpaired spins. Many organic radicals exist, but few are stable enough to be assembled into crystalline structures. Moreover, even when that is possible, *aligning* these spins ferromagnetically is usually impossible. Thus the discovery of ferromagnetism, albeit at rather low temperatures, in certain nitronyl nitroxide molecular crystals in the early 1990s was particularly remarkable. The first material of this sort to be found was *para*-nitrophenyl nitronyl nitroxide (*p*-NPNN) (see Fig. 8.26) which shows ferromagnetism up to  $T_C \sim 0.65$  K only in one of its crystal phases. Nitronyl nitroxides contain only the elements C, H, N, and O and are therefore fully **organic magnets**. On each nitronyl nitroxide molecule there is an unpaired spin associated with the two N–O groups. Small chemical changes to the rest of the molecule lead to significant changes in crystal structure, thereby altering the intermolecular overlaps and thus the magnetic interactions between unpaired spins on neighbouring molecules. Thus, different compounds have greatly different magnetic ground states. These materials mostly have transition temperatures below 1 K.

Some of the most technologically promising materials are **molecular magnets** in which a transition metal ion provides the localized moment and organic bridges act as exchange pathways. Progress has also been achieved using materials with unpaired electrons on both the metal ions and the organic molecule. These have much higher transition temperatures than purely organic magnets, sometimes above room temperature. One of the promising aspects of these materials is their optical properties. Some of them are transparent and change colour when they become magnetic, often when this is associated with a spin transition. They therefore have a number of potential applications in cases

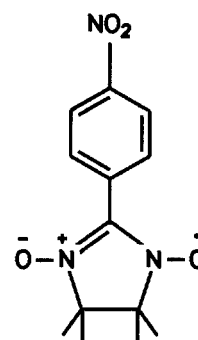


Fig. 8.26 The molecular structure of the organic ferromagnet *para*-nitrophenyl nitronyl nitroxide (*p*-NPNN).

when it is necessary to see by eye that a magnetic change has taken place (e.g. on telephone cards or display applications).

## 8.11 Spin electronics

The ability to manipulate and amplify currents of different spin-type is the basis of the emerging field of **spin electronics** (also known as **spintronics**) which is an attempt to fabricate devices out of combinations of metallic ferromagnets, non-magnetic metals and semiconductors. The ideas at the basis of this field are all connected with the fact that spin-up and spin-down electrons have different mobilities in a ferromagnet. A crucial feature is the degree of spin polarization in a given ferromagnet. The **spin-polarization**  $P$  is defined by

$$P = \left| \frac{n_{\uparrow} - n_{\downarrow}}{n_{\uparrow} + n_{\downarrow}} \right|, \quad (8.13)$$

and is less than unity for elemental metallic ferromagnets (0.44 for Fe, 0.34 for Co and 0.11 for Ni). However in some materials  $P = 1$ , implying that there is one spin-split band which is completely empty. Such materials are called **half-metallic magnets** (because the electrons of one spin are metallic and those of the other spin are insulating) and  $\text{CrO}_2$ ,  $\text{Fe}_3\text{O}_4$  and some of the manganites are believed to fall into this class. Such materials have a very long spin-diffusion length, and may be exciting materials for use in spin electronics applications, for example in **magnetic tunnel junctions** in which spin polarized electrons are used to tunnel through an insulating barrier.

A few different designs for a **spin transistor** have recently appeared which attempt to use the tunnelling of electrons of different spins from ferromagnets into semiconducting layers. Conventional electronics is based only on electronic charge and ignores electronic spin. There is much current optimism that the use of magnetic materials to control the spin of electronic currents could be a promising new development for electronic devices in this century.

## Further reading

- J. A. Mydosh, *Spin glasses: an experimental introduction*, Taylor and Francis 1993 and K. H. Fischer and J. A. Hertz, *Spin glasses*, CUP 1991, both contain useful accounts of theoretical and experimental work on spin glasses.
- E. Dagotto, *Reports on Progress in Physics* **62**, 1525 (1999) is a readable review on spin ladders.
- A. M. Tsvelik, *Quantum field theory in condensed matter physics*, CUP 1995, is an advanced text which contains a description of current theories used in describing low-dimensional magnetism.
- A. Auerbach, *Interacting electrons and quantum magnetism*, Springer-Verlag 1994, is a lively treatment of the quantum mechanics of low-dimensional magnets and the Haldane gap. These topics are also reviewed in I. Affleck, *J. Phys.: Condensed Matter*, **1**, 3047 (1989).
- M. A. Kastner, R. J. Birgeneau, G. Shirane and Y. Endoh, *Rev. Mod. Phys.*, **70**, 897 (1998) contains a review of magnetic properties of single layer copper oxides and the high- $T_c$  problem.
- E. Manousakis, *Rev. Mod. Phys.*, **63**, 1 (1991) reviews the theory of the two-dimensional Heisenberg antiferromagnet on a square lattice.
- P. M. Chaikin and T. C. Lubensky, *Principles of condensed matter physics*, CUP 1995, is an excellent source of information concerning the statistical mechanics of systems in condensed matter.

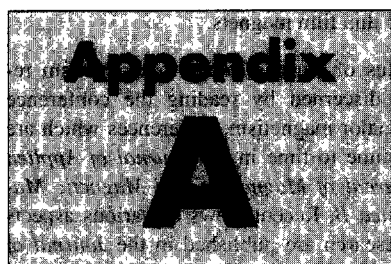
- Quantum phase transitions are reviewed in S. L. Sondhi, S. M. Girvin, J. P. Carini and D. Shahar, *Rev. Mod. Phys.*, **69**, 315 (1997).
- A review of magnetic multilayers with an emphasis on spin-wave excitations is R. E. Camley and R. L. Stamps, *J. Phys.: Condensed Matter*, **5**, 3727 (1993).
- Organic magnetism is reviewed in O. Kahn, *Molecular magnetism*, VCH (1993).
- K. De'Bell, A. B. MacIsaac and J. P. Whitehead, *Rev. Mod. Phys.*, **72**, 225 (2000) contains an account of

dipolar effects in thin film magnets.

- The current status of various topics in magnetism research may be discerned by reading the conference proceedings of major magnetism conferences which are published from time to time in the *Journal of Applied Physics*, the *Journal of Magnetism and Magnetic Materials* and *Physica B*. Recent views of various aspects of magnetism research are published in the *Journal of Magnetism and Magnetic Materials* volume  $100 \times n$  where  $n$  is an integer.

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# Units in electromagnetism

Units in electromagnetism represent a potential minefield of complication. SI units are used throughout this book, but much of the twentieth century literature uses non-SI notation. Therefore this appendix attempts to unravel some of the mysteries of the old system and it may be used to interpret older literature for readers who have been brought up on the new system.

From the end of the nineteenth century until just after the second world war, the **cgs system** was the system of units preferred by the international committees who are concerned with such things. In the cgs system, distance is measured in centimetres, mass in grams, and time in seconds, hence the name cgs. In the **SI system**, we use metres (100 times as big as centimetres), kilograms (1000 times as big as grams) and seconds. This difference in choice of units of distance and mass appears to be rather innocent, but ends up causing great changes in other units. For example, force has the dimensions of mass  $\times$  length / time<sup>2</sup>, so in cgs the unit of force, the dyne, is  $1000 \times 100 = 10^5$  times smaller than a Newton, the SI unit of force. The cgs unit of energy is the erg, a dyne cm, so this is  $10^5 \times 10^2 = 10^7$  times smaller than the Joule, the SI unit of energy.

Unfortunately, that is not the end of it, and when it comes to the definitions of magnetic field things get decidedly worse. This arises because the choice of definitions of the fields **B**, **H** and **M** are different in the two systems:

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) \quad (\text{SI}) \quad \mathbf{B} = \mathbf{H} + 4\pi\mathbf{M} \quad (\text{cgs}) \quad (\text{A.1})$$

Thus **B** and **H** have the same dimensions in the cgs system but confusingly have different units (the former in Gauss (G), the latter in Oersted (Oe)). The difference in relation between **H** and **M** arises because the two systems make different choices in the equations defining the magnitude of the magnetic field, **H**, at a distance *r* from a magnetic monopole *q*:

$$H = \frac{q}{4\pi r^2} \quad (\text{SI}) \quad H = \frac{q}{r^2} \quad (\text{cgs}) \quad (\text{A.2})$$

The omission of the factor of  $4\pi$  makes this equation, and similar ones relating the magnetic field from a magnetic dipole, rather simpler in the cgs system. However the factor of  $4\pi$  is not removed entirely from the theory but pops up in a more unwanted place in eqn A.1. The cgs system defined **H** according to non-existent monopoles from eqn A.2, so that a unit monopole produces a field of one Oersted a distance of one centimetre away (and if another unit monopole is lurking there it will experience a force of one dyne). In SI, **H** is more helpfully defined in terms of current (since people normally use a current source in a laboratory to produce a magnetic field, rather than devising

a suitable arrangement of unit magnetic monopoles). Thus a long solenoid with  $n$  turns per metre and carrying a current of  $I$  Amps produces a field given by  $H = nI$  in the SI system.

The logic in the SI system is that the  $4\pi$  appears in equations like eqn A.2 to remind us that the field from a point magnetic monopole has spherical symmetry (the  $4\pi$  is related to the surface area of a sphere,  $4\pi r^2$ ). The same is true of the electric field  $E$  a distance  $r$  from a point charge,

$$E = \frac{q}{4\pi\epsilon_0 r^2} \quad (\text{SI}) \quad E = \frac{q}{r^2} \quad (\text{cgs}), \quad (\text{A.3})$$

in which similar arguments apply. With SI, we have to live with a few extra factors, which fit from symmetry arguments, in equations defining the field from point charges and dipoles. The rest of the equations come out much cleaner and with fewer arbitrary factors.

The one thing to be said in favour of the cgs system is that electric and magnetic fields come out with the same dimensions, so that certain relativistic equations can look better. Also in free space  $\mathbf{B} = \mathbf{H}$  which has a certain symmetry (although this can create an opportunity for confusion). But extra factors of  $c$  get littered all over the place in the cgs system, so that for example the equation for the Lorentz force on a charge moving with velocity  $\mathbf{v}$  in electric and magnetic fields becomes

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (\text{SI}) \quad \mathbf{F} = q \left( \mathbf{E} + \frac{\mathbf{v} \times \mathbf{B}}{c} \right) \quad (\text{cgs}). \quad (\text{A.4})$$

But there is much worse to come. It turns out that there are two ways to set up electromagnetism in cgs, with either the esu (electrostatic units) or the emu (electromagnetic units) with conversion factors needed to go between them. In magnetism the emu system is usually adopted. The emu unit of magnetic moment, usually abbreviated just to the 'emu', is a unit with dimensions of volume (which in cgs is  $\text{cm}^3$ ). Magnetic susceptibility,  $\chi = M/H$ , is dimensionless in both systems, but in cgs is often written as  $\text{emu cm}^{-3}$ , which is still dimensionless even if it doesn't appear so at first glance. There is always the troublesome factor of  $4\pi$  to remember when converting between cgs susceptibility and SI susceptibility which arises because  $\mathbf{B} = \mathbf{H} + 4\pi\mathbf{M}$  in the cgs system (see above). For example, the demagnetization factor,  $N$ , is dimensionless and is  $0 < N < 1$  in SI but  $0 < N < 4\pi$  in cgs. Unfortunately the emu is still widely used in some parts of magnetism research. Table A summarizes the most useful results for converting SI to and from cgs.

Though not an SI unit, the Bohr magneton  $\mu_B = 9.274 \times 10^{-24} \text{ J T}^{-1}$  is a useful measure of magnetic moment since it corresponds to the magnetic moment of a 1s electron in hydrogen. For a paramagnet, the molar susceptibility  $\chi_m$  is given by Curie's law (adapting eqn 2.44) which is in SI units

$$\chi_m = \frac{\mu_0 N_A \mu_{\text{eff}}^2 \mu_B^2}{3k_B T} \quad (\text{A.5})$$

where  $N_A$  is Avogadro's number. Hence  $\chi_m T$  is independent of temperature (see Fig. 2.10) and this can be related to the effective moment. Hence by rearranging eqn A.5, one has

$$\mu_{\text{eff}} = [3k_B / \mu_0 N_A \mu_B^2]^{1/2} \sqrt{\chi_m T}, \quad (\text{A.6})$$

**Table A.1** Units in the SI system and the cgs system. The abbreviations are m = metre, g = gramme, N = Newton, J = Joule, T = Tesla, G = Gauss, A = Amp, Oe = Oersted, Wb = Weber, Mx = Maxwell. The term emu is short for electromagnetic unit. Note that magnetic susceptibility is dimensionless in SI units.

| Quantity                               | symbol       |                       | SI unit                                                   |     | cgs unit              |
|----------------------------------------|--------------|-----------------------|-----------------------------------------------------------|-----|-----------------------|
| Length                                 | $x$          | $10^{-2}$             | m                                                         | = 1 | cm                    |
| Mass                                   | $m$          | $10^{-3}$             | kg                                                        | = 1 | g                     |
| Force                                  | $F$          | $10^{-5}$             | N                                                         | = 1 | dyne                  |
| Energy                                 | $E$          | $10^{-7}$             | J                                                         | = 1 | erg                   |
| Magnetic induction                     | $\mathbf{B}$ | $10^{-4}$             | T                                                         | = 1 | G                     |
| Magnetic field strength                | $\mathbf{H}$ | $10^3/4\pi$           | A m <sup>-1</sup>                                         | = 1 | Oe                    |
| Magnetic moment                        | $\mu$        | $10^{-3}$             | J T <sup>-1</sup>                                         | = 1 | erg G <sup>-1</sup>   |
|                                        |              |                       | or A m <sup>2</sup>                                       |     |                       |
| Magnetization<br>(= moment per volume) | $\mathbf{M}$ | $10^3$                | A m <sup>-1</sup> or<br>J T <sup>-1</sup> m <sup>-3</sup> | = 1 | Oe                    |
| Magnetic susceptibility                | $\chi$       | $4\pi$                |                                                           | = 1 | emu cm <sup>-3</sup>  |
| Molar susceptibility                   | $\chi_m$     | $4\pi \times 10^{-6}$ | m <sup>3</sup> mol <sup>-1</sup>                          | = 1 | emu mol <sup>-1</sup> |
| Mass susceptibility                    | $\chi_g$     | $4\pi \times 10^{-3}$ | m <sup>3</sup> kg <sup>-1</sup>                           | = 1 | emu g <sup>-1</sup>   |
| Magnetic flux                          | $\phi$       | $10^{-8}$             | T m <sup>2</sup>                                          | = 1 | G cm <sup>-2</sup>    |
|                                        |              |                       | or Wb                                                     |     | or Mx                 |
| Demagnetization factor                 | $N$          |                       | $0 < N < 1$                                               |     | $0 < N < 4\pi$        |

so that

$$\mu_{\text{eff}} = 797.8 \sqrt{\chi_m^{\text{SI}} T} \quad (\text{SI}) \quad (\text{A.7})$$

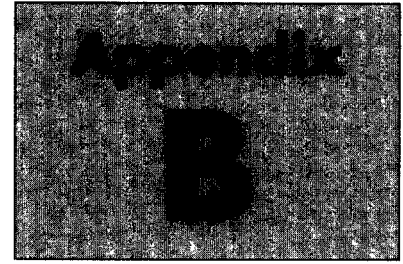
$$\mu_{\text{eff}} = 2.827 \sqrt{\chi_m^{\text{cgs}} T} \quad (\text{cgs}) \quad (\text{A.8})$$

where  $\mu_{\text{eff}}$  is measured in Bohr magnetons per formula unit,  $\chi_m^{\text{SI}}$  is measured in m<sup>3</sup> mol<sup>-1</sup>, and  $\chi_m^{\text{cgs}}$  is measured in emu mol<sup>-1</sup>. These numerical relationships can be useful for extracting effective moments from graphs of  $\chi_m T$  against  $T$ .

## Further reading

- B. I. Bleaney and B. Bleaney, *Electricity and magnetism*, OUP 1989.
- J. D. Jackson, *Classical electrodynamics*, Wiley 1962.

# Electromagnetism



In this appendix several key results in electromagnetism are briefly reviewed. The further reading should be consulted for detailed derivations of these results.

## B.1 Magnetic moments

The Lorentz force  $d\mathbf{F}$  on a straight segment of wire of length  $dr$  carrying current  $I$  in a magnetic field of flux density  $\mathbf{B}$  is given by  $d\mathbf{F} = I d\mathbf{r} \times \mathbf{B}$  (see Fig. B.1).

This result can now be applied to calculate the couple on the elementary loop of wire in Fig. B.2. The couple is given by

$$\begin{aligned} dG_x &= -I dx dy B_y \\ dG_y &= I dx dy B_x, \end{aligned} \quad (\text{B.1})$$

so that defining the magnetic moment of the loop by  $d\mu_z = I dx dy$ , and repeating the argument for arbitrary directions of the loop, the couple  $d\mathbf{G}$  is given by

$$d\mathbf{G} = d\boldsymbol{\mu} \times \mathbf{B}. \quad (\text{B.2})$$

For a loop of finite size, this yields by integration

$$\mathbf{G} = \boldsymbol{\mu} \times \mathbf{B}. \quad (\text{B.3})$$

There will be a net force on the loop only if the magnetic field is non-uniform. Thus the force in the  $x$  direction on the elementary loop in Fig. B.2 is

$$dF_x = I dy \left( B_z + \frac{\partial B_z}{\partial x} dx \right) - I dy B_z = d\mu_z \frac{\partial B_z}{\partial x}, \quad (\text{B.4})$$

so that

$$d\mathbf{F} = (d\boldsymbol{\mu} \cdot \nabla) \mathbf{B} \quad (\text{B.5})$$

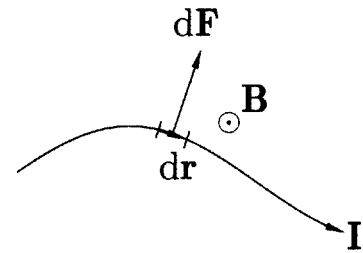
and by integrating for a magnetic moment of finite size,

$$\mathbf{F} = (\boldsymbol{\mu} \cdot \nabla) \mathbf{B}. \quad (\text{B.6})$$

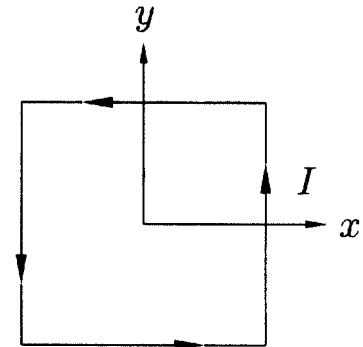
It is therefore only a non-uniform magnetic field which can exert a force on a magnetic moment.

A magnetic moment  $\boldsymbol{\mu}$  aligned with a magnetic field  $\mathbf{B}$  has zero torque on it (by eqn B.3,  $\mathbf{G} = 0$  if  $\boldsymbol{\mu}$  and  $\mathbf{B}$  are parallel). Rotating  $\boldsymbol{\mu}$  with respect to  $\mathbf{B}$

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**Fig. B.1** The force on a current element is  $d\mathbf{F} = I d\mathbf{r} \times \mathbf{B}$ .



**Fig. B.2** A square elementary current loop (with sides of length  $dx$  and  $dy$ ) oriented perpendicular to the  $z$ -direction.

therefore can only be accomplished by applying a torque. The work done  $W$  by the torque to rotate  $\mu$  to an angle  $\theta$  with respect to the magnetic field is

$$W = \int_0^\theta G d\theta' = \int_0^\theta \mu B \sin \theta' d\theta' = \mu B(1 - \cos \theta). \quad (\text{B.7})$$

The potential energy  $U$  of the magnetic moment can therefore be written as  $-\mu B \cos \theta$  (ignoring the constant term). Hence we can write

$$U = -\mu \cdot \mathbf{B}. \quad (\text{B.8})$$

A magnetic moment  $\mu$  in a non-uniform field  $\mathbf{B}$  has a force on it in the  $x$  direction equal to  $\mu \partial B / \partial x$  by eqn B.6. To stop it moving in the  $x$  direction, one would have to apply a restoring force  $-\mu \partial B / \partial x$ . If it were allowed to move a distance  $\delta x$  into a region of stronger field, the work done by this force would be

$$dW = -\mu \frac{\partial B}{\partial x} \delta x = -\mu \delta B, \quad (\text{B.9})$$

and this is an increment to the sample's free energy (in agreement with the results of Exercise 2.10).

The magnetic field due to a magnetic moment at the origin is

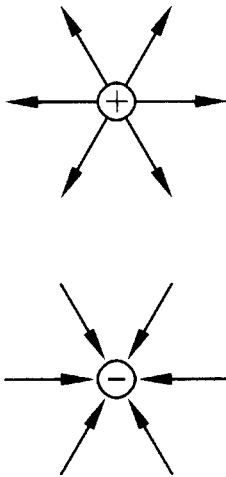
$$\mathbf{B}(\mathbf{r}) = -\mu_0 \nabla \left( \frac{\mu \cdot \mathbf{r}}{4\pi r^3} \right) \quad (\text{B.10})$$

and the magnetic vector potential (in the most convenient choice of gauge) is

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{\mu \times \mathbf{r}}{r^3}. \quad (\text{B.11})$$

## B.2 Maxwell's equations in free space

James Clerk Maxwell (1831–1879)



**Fig. B.3** Electric field diverges away from positive charges and converges into negative charges.

Maxwell's equations in free space are

$$\nabla \cdot \mathbf{E} = \rho / \epsilon_0 \quad (\text{B.12})$$

$$\nabla \cdot \mathbf{B} = 0 \quad (\text{B.13})$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (\text{B.14})$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \epsilon_0 \mu_0 \frac{\partial \mathbf{E}}{\partial t}, \quad (\text{B.15})$$

and describe the interrelationships between the electric field  $\mathbf{E}$ , the magnetic induction  $\mathbf{B}$ , the charge density  $\rho$  and the current density  $\mathbf{J}$ . Equation B.12 shows that electric field diverges away from positive charges and converges into negative charges; charge density therefore acts as a source or a sink of electric field (see Fig. B.3). Equation B.13 shows that magnetic fields have no such divergence; thus there are no magnetic charges (monopoles) and lines of  $\mathbf{B}$  field must just exist in loops; they can never start or stop anywhere. Equation B.14 shows that you only get loops of electric field around regions in space in which there is a changing magnetic field. This leads to Faraday's law of electromagnetic induction. Equation B.15, in the absence of a changing electric field, shows that loops of magnetic induction are found around electric currents. Note that everything in this section has been for electromagnetic fields in free space.

### B.3 Free and bound currents

Once matter is brought into the picture, things get a little more complicated and it is worth trying to understand why. The treatment here will just consider the effects for magnetism, although analogous thinking lies behind the corresponding derivations for electric fields. Real matter contains a different type of electric current from the familiar currents we are used to thinking about flowing down wires. Atoms contain microscopic **bound currents** due to the electrons circulating around the nucleus. Thus the total current  $\mathbf{J}$  in eqn B.15 can be separated into two components

$$\mathbf{J} = \mathbf{J}_{\text{free}} + \mathbf{J}_{\text{bound}}, \quad (\text{B.16})$$

due to the free currents (10 amps flowing down a copper wire) and bound currents (electrons orbiting an atom).

If  $\mathbf{M}$  is uniform in a sample, the only net bound current that flows is around the edge of the material. If  $\mathbf{M}$  is non-uniform, bound currents will be generated inside the material. We need a general relation between the bound currents and the magnetization, and this is that

$$\mathbf{J}_{\text{bound}} = \nabla \times \mathbf{M}. \quad (\text{B.17})$$

A derivation of eqn B.17 will now be sketched (for more details see the further reading). The vector potential  $\mathbf{A}$  due to a single point dipole at the origin is given in eqn B.11. Now consider a magnetized specimen, contained in volume  $V$ , with magnetization  $\mathbf{M}(\mathbf{r}')$  at position  $\mathbf{r}'$ . The vector potential at position  $\mathbf{r}$  is

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_V d^3r' \frac{\mathbf{M}(\mathbf{r}') \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3}. \quad (\text{B.18})$$

This equation can be simplified using  $\nabla_r(1/r) = -\mathbf{r}/r^3$ , the vector identity

$$\nabla \times (f\mathbf{g}) = f\nabla \times \mathbf{g} - \mathbf{g} \times \nabla f, \quad (\text{B.19})$$

where  $f$  and  $\mathbf{g}$  are scalar and vector functions respectively, and transforming the integral of  $\mathbf{r}'$  to one over  $\mathbf{r}$ . The result is that the vector potential at  $\mathbf{r}$  can be written as

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_V \frac{\nabla \times \mathbf{M} d\tau}{r} + \frac{\mu_0}{4\pi} \int_S \nabla \times \left( \frac{\mathbf{M}}{r} \right) d\tau \quad (\text{B.20})$$

$$= \frac{\mu_0}{4\pi} \int_V \frac{\nabla \times \mathbf{M} d\tau}{r} + \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{M} \times d\mathbf{S}}{r} \quad (\text{B.21})$$

$$= \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{J}_{\text{bound}} d\tau}{r} + \frac{\mu_0}{4\pi} \int_S \frac{\mathbf{K}_{\text{bound}} dS}{r} \quad (\text{B.22})$$

where the first integral is over the volume of the sample, and the second integral is over the surface of the sample. The bound current in eqn B.22 agrees with eqn B.17. The surface current is given by  $\mathbf{K}_{\text{bound}} = \mathbf{M} \times \hat{\mathbf{n}}$  and the vector  $\hat{\mathbf{n}}$  is perpendicular to the surface.

The problem with eqn B.15 is that the curl of  $\mathbf{B}$  is related not only to the free current density but also to the more inconvenient bound current density. Therefore we are motivated to define a new ‘corrected’ field which only ‘sees’

the free current density, although as shown below this will be only partially successful. Using the result for the bound volume currents, we define a new field  $\mathbf{H}$  (called the **magnetic field strength**, or just the  $H$  field) by

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) \quad (\text{B.23})$$

and hence we can write a new version of eqn B.15 which includes only  $\mathbf{J}_{\text{free}}$ . Thus in the absence of changing electric fields

$$\nabla \times \mathbf{H} = \mathbf{J}_{\text{free}}. \quad (\text{B.24})$$

This is useful because it relates the  $\mathbf{H}$ -field to a quantity that can be read on an ammeter (which  $\mathbf{J}_{\text{bound}}$  cannot be). Unfortunately this procedure, having simplified one of Maxwell's equation, complicates another. Equation B.13 changes from the simple  $\nabla \cdot \mathbf{B} = 0$  to the slightly more tortuous

$$\nabla \cdot \mathbf{H} = -\nabla \cdot \mathbf{M}. \quad (\text{B.25})$$

Thus  $\mathbf{H}$ , unlike  $\mathbf{B}$ , is not divergence-free (or **solenoidal**, to use the mathematical jargon).  $\mathbf{H}$  behaves as if magnetic monopoles (i.e. the magnetic equivalent of charges) exist.

## B.4 Maxwell's equations in matter

Maxwell's equations in the presence of matter therefore become

$$\nabla \cdot \mathbf{D} = \rho_{\text{free}} \quad (\text{B.26})$$

$$\nabla \cdot \mathbf{B} = 0 \quad (\text{B.27})$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (\text{B.28})$$

$$\nabla \times \mathbf{H} = \mathbf{J}_{\text{free}} + \frac{\partial \mathbf{D}}{\partial t}, \quad (\text{B.29})$$

where a similar distinction is made between free and bound charges and where the **electric displacement**,  $\mathbf{D}$ , is given by

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} \quad (\text{B.30})$$

and  $\mathbf{P}$  is the electric polarization.

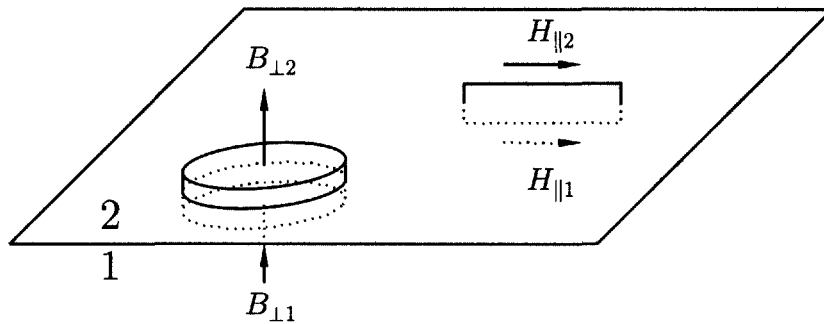
## B.5 Boundary conditions

<sup>1</sup>These are traditionally called 'pill-boxes' in the physics literature, although pills have long since stopped being sold in such shaped boxes!

<sup>2</sup>And we can make sure this is the case by making sure that we let the box thickness tend to zero before its radius tends to zero.

A significant difference between the  $\mathbf{B}$  and  $\mathbf{H}$  fields is revealed when they meet a boundary between two different regions. The boundary conditions between two regions 1 and 2 (see Fig. B.4) can be derived as follows. Consider the shoe-polish-tin-shaped box<sup>1</sup> straddling the surface.  $\nabla \cdot \mathbf{B} = 0$  implies that  $\int_S \mathbf{B} \cdot d\mathbf{S} = 0$  using the divergence theorem. Since the surface area of the box is mostly represented by the top and bottom surfaces of the box,<sup>2</sup> we have that the perpendicular component of  $\mathbf{B}$  is continuous, i.e. that

$$B_{\perp 1} = B_{\perp 2}. \quad (\text{B.31})$$



**Fig. B.4** Electromagnetic boundary conditions between regions 1 and 2.

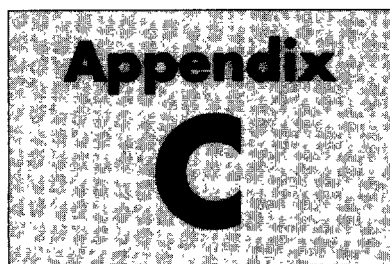
The argument won't work for  $\mathbf{H}$  because  $\nabla \cdot \mathbf{H} = -\nabla \cdot \mathbf{M}$  which may be non-zero. The perpendicular component may be discontinuous if the magnetization diverges inside the box (which it may well do if region 1 is magnetized and region 2 is not). The argument for the parallel components comes from the equation  $\nabla \times \mathbf{H} = \mathbf{J}_{\text{free}}$  which implies that  $\oint \mathbf{H} \cdot d\mathbf{l} = \int \mathbf{J}_{\text{free}} \cdot d\mathbf{S}$ . As the Amperean loop shown on the right-hand side of Fig. B.4 shrinks to zero, so does the free current enclosed by the loop and so the parallel component of  $\mathbf{H}$  is continuous, i.e. that

$$H_{\parallel 1} = H_{\parallel 2}. \quad (\text{B.32})$$

This argument won't work for  $\mathbf{B}$  because  $\nabla \times \mathbf{B} = \mu_0(\mathbf{J}_{\text{free}} + \mathbf{J}_{\text{bound}})$  and so the Amperean loop, although enclosing no free current as it shrinks to zero, can enclose some bound surface current in the limit that the loop shrinks to zero.

## Further reading

- B. I. Bleaney and B. Bleaney, *Electricity and magnetism*, OUP 1989.
- D. J. Griffiths, *Introduction to electrodynamics*, Prentice-Hall 1989.
- J. D. Jackson, *Classical electrodynamics*, John-Wiley 1962.



# Quantum and atomic physics

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## C.1 Quantum mechanics

In this section we briefly review some basic features of quantum mechanics. The fundamental object in quantum mechanics is the **wave function** (also known as the **state function**)  $\psi$  which contains all the knowledge one obtains about a system from observation. The wave function  $\psi$  is a complex function and  $|\psi|^2 = \psi\psi^*$  is the probability density.<sup>1</sup> To every observable quantity, there corresponds a Hermitian operator. When you make a measurement, the result you obtain is one of the eigenvalues of that operator. Hermitian operators have real eigenvalues and eigenfunctions which are orthogonal to each other. If the  $i^{\text{th}}$  eigenfunction of the operator  $\hat{A}$  (operators are given 'hats') is  $\phi_i$  (assumed normalized) and has an eigenvalue  $a_i$ , then

$$\hat{A}\phi_i = a_i\phi_i. \quad (\text{C.1})$$

The expected value that you obtain after a measurement of operator  $\hat{A}$  is given by

$$\langle \hat{A} \rangle = \int d\tau \psi^* \hat{A} \psi, \quad (\text{C.2})$$

where  $d\tau$  is a volume element. The wave function  $\psi$  can be expanded in terms of the eigenfunctions of  $\hat{A}$ , i.e.

$$\psi = \sum_i c_i \phi_i, \quad (\text{C.3})$$

so that  $\langle \hat{A} \rangle$  is then given by

$$\langle \hat{A} \rangle = \sum_i |c_i|^2 a_i, \quad (\text{C.4})$$

using the orthonormality of  $\phi_i$ . The probability of obtaining the  $i^{\text{th}}$  eigenvalue,  $a_i$ , is  $|c_i|^2$ . Measurement is a drastic process. If you take a wave function  $\psi$ , measure the physical quantity which is associated with operator  $\hat{A}$  and obtain the result  $a_n$  (the  $n^{\text{th}}$  eigenvalue of  $\hat{A}$ ), the system is forced into state function  $\phi_n$ . This, at least, is the Copenhagen interpretation.

The commutator of two operators  $\hat{A}$  and  $\hat{B}$  is defined by  $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$ . An expression defining a commutator is known as a **commutation relation**.

<sup>1</sup> Here the symbol \* denotes complex conjugation.

**Example C.1**

The commutator  $[\hat{x}, \hat{p}] = i\hbar$  where  $\hat{x}$  and  $\hat{p}$  are the position and momentum operators respectively.

If the two operators  $\hat{A}$  and  $\hat{B}$  commute, ( $[\hat{A}, \hat{B}] = 0$ ), they are said to be **compatible** and measurement of one does not affect the value of the other in any way. If they do not commute, there exists an uncertainty relation between them so that

$$\Delta A \Delta B \geq \frac{1}{2} |\langle i[\hat{A}, \hat{B}] \rangle|, \quad (\text{C.5})$$

where  $\Delta A \equiv \sqrt{\langle (\hat{A} - \langle \hat{A} \rangle)^2 \rangle}$ .

The time dependence of  $\psi$  in the absence of a measurement is given by the Schrödinger equation

$$\hat{\mathcal{H}}\psi = i\hbar \frac{d\psi}{dt} \quad (\text{C.6})$$

where  $\hat{\mathcal{H}}$  is the Hamiltonian. The time dependence of the expected value of an operator is given by

$$\frac{d\langle \hat{A} \rangle}{dt} = \frac{1}{i\hbar} \langle [\hat{A}, \hat{\mathcal{H}}] \rangle, \quad (\text{C.7})$$

if the operator  $\hat{A}$  itself is not time dependent. An observable whose operator commutes with the Hamiltonian is a conserved quantity and is known as a **constant of the motion** or a **good quantum number**.

## C.2 Dirac bra and ket notation

In this section we review some notation, due to Paul Dirac, which is used sporadically in this text and often in the magnetism literature. The scalar product between two vectors  $\mathbf{a}$  and  $\mathbf{b}$  is given by  $\mathbf{a} \cdot \mathbf{b} = \sum_i a_i b_i$ . The length of  $\mathbf{a}$  is given by  $\sqrt{\mathbf{a} \cdot \mathbf{a}}$  which must be real and positive. The notation makes the inner product look completely symmetric between  $\mathbf{a}$  and  $\mathbf{b}$ , but this hides some subtlety. If the vectors are complex, the scalar product should be written

Paul A. M. Dirac (1902–1984)

$$\mathbf{a}^\dagger \cdot \mathbf{b} = \sum_i a_i^* b_i = (a_1^* \ a_2^* \ a_3^*) \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}. \quad (\text{C.8})$$

This definition ensures that  $\mathbf{a}^\dagger \cdot \mathbf{a} = \sum_i |a_i|^2$  is real and positive, and so the length of the complex vector is well defined. Equation C.8 emphasizes that the two vectors in the scalar product should not really be considered on an equal footing: one is a row vector, the other a column vector; one is complex conjugated, the other is not. Mathematicians say that actually the two vectors in a scalar product ‘live’ in different spaces, so that here  $\mathbf{b}$  lives in a vector space and  $\mathbf{a}$  lives in its ‘dual space’. To turn a vector into its dual, you need

to take the adjoint, signified by a  $\dagger$  sign (it means ‘complex conjugate and transpose’), so that

$$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}^\dagger = (a_1^* \ a_2^* \ a_3^*). \quad (\text{C.9})$$

However in quantum mechanics, the state function is sometimes a vector, like  $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ , and sometimes a continuous function  $a(x)$ . If a function, then the scalar product between  $a(x)$  and  $b(x)$  is

$$a^\dagger \cdot b = \int dx \ a^*(x)b(x). \quad (\text{C.10})$$

Dirac circumvented this notational problem by writing the state function as a **ket**:  $|a\rangle$ , whether he was dealing with finite-component vectors or functions (which can be regarded as infinite dimensional vectors). This is *just* notation, so that  $|a\rangle + |b\rangle = |c\rangle$  is simply another way of writing  $\mathbf{a} + \mathbf{b} = \mathbf{c}$ , or equivalently  $\vec{a} + \vec{b} = \vec{c}$  or  $a(x) + b(x) = c(x)$ . One can turn the ket  $|a\rangle$  into a **bra**,  $\langle a|$ , by using the adjoint

$$|a\rangle^\dagger = \langle a|. \quad (\text{C.11})$$

The scalar product between  $\langle a|$  and  $|b\rangle$  is then written as

$$\langle a|b\rangle = \sum_i a_i^* b_i = \int dx \ a^*(x)b(x), \quad (\text{C.12})$$

a bra-(c)-ket (Dirac here demonstrating evidence of humour). The expression for an expectation value of an operator can be written

$$\langle \hat{A} \rangle = \langle \psi | \hat{A} | \psi \rangle. \quad (\text{C.13})$$

The notation conveys the asymmetry of the scalar product, is widely used and greatly simplifies many computations. It is particularly useful in labelling particular states, as in the following:

$$|\text{Schrödinger's cat}\rangle = \frac{|\text{alive}\rangle + |\text{dead}\rangle}{\sqrt{2}}. \quad (\text{C.14})$$

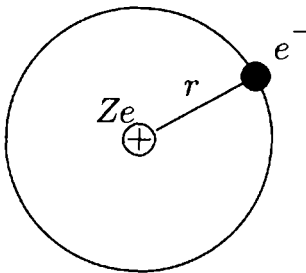


Fig. C.1 The Bohr model.

### C.3 The Bohr model

The Bohr model is a semiclassical model describing the motion of an electron in a hydrogen atom. It gives the right results, but for the wrong reason. Nevertheless, it is simple and therefore useful for quickly deriving results such as the dependence of the radius of an atomic orbit on nuclear charge  $Z$  and effective mass.

Consider an electron orbiting a nucleus of charge  $+Ze$  (see Fig. C.1). Equating electrostatic and centripetal forces yields

$$\frac{Ze^2}{4\pi\epsilon_0 r^2} = \frac{m_e v^2}{r}. \quad (\text{C.15})$$

The angular momentum of the electron  $m_e v r$  is quantized in units of  $\hbar$  so we can write

$$m_e v r = n \hbar, \quad (\text{C.16})$$

where  $n$  is an integer (the principal quantum number). Hence the radius of the orbit can be written

$$r = \frac{4\pi\epsilon_0\hbar^2 n^2}{Ze^2 m_e} = \left( \frac{4\pi\epsilon_0\hbar^2}{m_e e^2} \right) \frac{n^2}{Z} = \frac{a_0 n^2}{Z}, \quad (\text{C.17})$$

where  $a_0$  is the Bohr radius defined by

$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2}. \quad (\text{C.18})$$

The energy  $E$  is then given by

$$E = \frac{1}{2} m v^2 - \frac{Z e^2}{4\pi\epsilon_0 r} \quad (\text{C.19})$$

$$= -\frac{1}{2} \left( \frac{Z e^2}{4\pi\epsilon_0 r} \right) \quad (\text{C.20})$$

$$= -\frac{\hbar^2}{2m_e} \frac{1}{a_0^2} \frac{Z^2}{n^2} \quad (\text{C.21})$$

$$= -\frac{m_e e^4}{(4\pi\epsilon_0\hbar)^2} \frac{Z^2}{n^2}. \quad (\text{C.22})$$

These results can be written as  $a_0 = \frac{\hbar}{m\alpha c}$  and  $E = -\frac{1}{2} m(\alpha c)^2 \frac{Z^2}{n^2}$  where  $\alpha$  is the fine structure constant.

## C.4 Orbital angular momentum

The angular momentum operator is given by

$$\hbar \hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}} = -i\hbar \hat{\mathbf{r}} \times \nabla. \quad (\text{C.23})$$

The  $z$  component of the angular momentum thus has an operator

$$\hbar \hat{L}_z = i\hbar \left[ y \frac{\partial}{\partial x} - x \frac{\partial}{\partial y} \right] \quad (\text{C.24})$$

$$= -i\hbar \frac{\partial}{\partial \phi} \quad (\text{C.25})$$

which has eigenfunctions given by  $e^{im_l \phi}$  and eigenvalues  $m_l \hbar$ , where  $m_l$  is the magnetic quantum number. The following commutation relations can be easily proved:

$$[\hat{L}_i, \hat{L}^2] = 0 \quad (\text{C.26})$$

and

$$[\hat{L}_i, \hat{L}_j] = i\epsilon_{ijk} \hat{L}_k \quad (\text{C.27})$$

where the alternating tensor  $\epsilon_{ijk}$  is defined by

$$\epsilon_{ijk} = \begin{cases} 1 & \text{if } ijk \text{ is an even permutation of } 123 \\ -1 & \text{if } ijk \text{ is an odd permutation of } 123 \\ 0 & \text{if any two of } i, j \text{ or } k \text{ are equal.} \end{cases} \quad (\text{C.28})$$

**Table C.1**  $P_l^{m_l}(\cos \theta)$ .

|         | $m_l = 0$                                      | $m_l = 1$                                       | $m_l = 2$                                | $m_l = 3$                    |
|---------|------------------------------------------------|-------------------------------------------------|------------------------------------------|------------------------------|
| $l = 0$ | 1                                              | —                                               | —                                        | —                            |
| $l = 1$ | $\cos \theta$                                  | $\sin \theta$                                   | —                                        | —                            |
| $l = 2$ | $\frac{1}{2}(3 \cos^2 \theta - 1)$             | $\frac{1}{3} \sin \theta \cos \theta$           | $\frac{1}{3} \sin^2 \theta$              | —                            |
| $l = 3$ | $\frac{1}{2}(5 \cos^3 \theta - 3 \cos \theta)$ | $\frac{2}{3} \sin \theta (5 \cos^2 \theta - 1)$ | $\frac{1}{15} \sin^2 \theta \cos \theta$ | $\frac{1}{15} \sin^3 \theta$ |

Equation C.27 is shorthand for

$$[\hat{L}_x, \hat{L}_y] = i\hat{L}_z$$

and cyclic permutations. The equation uses Einstein's summation convention in which any twice-repeated index is assumed summed. For example  $a_i b_i$  is short for  $\sum_i a_i b_i$ . The alternating tensor is useful in expressions such as

$$(\mathbf{a} \times \mathbf{b})_i = \epsilon_{ijk} a_j b_k.$$

In this equation  $j$  and  $k$  are twice-repeated and thus assumed summed. Hence for example

$$\begin{aligned} (\mathbf{a} \times \mathbf{b})_1 &= \epsilon_{123} a_2 b_3 + \epsilon_{132} a_3 b_2 \\ &= a_2 b_3 - a_3 b_2. \end{aligned}$$

The operator  $\hat{L}^2$  has eigenfunctions

$$|l, m_l\rangle = Y_{lm_l}(\theta, \phi) \propto P_l^{m_l}(\cos \theta) e^{im_l \phi}, \quad (\text{C.29})$$

known as spherical harmonics, with eigenvalues  $l(l+1)$  where  $l$  is the angular momentum quantum number and  $P_l^{m_l}(\cos \theta)$  is an associated Legendre polynomial (see Table C.1).

Thus

$$\hat{L}^2 |l, m_l\rangle = l(l+1) |l, m_l\rangle. \quad (\text{C.30})$$

Similarly

$$\hat{L}_z |l, m_l\rangle = m_l |l, m_l\rangle. \quad (\text{C.31})$$

The raising and lowering operators  $\hat{L}_{\pm}$  are defined by

$$\hat{L}_{\pm} = \hat{L}_x \pm i\hat{L}_y. \quad (\text{C.32})$$

Hence one can show that

$$\hat{L}_{\pm} |l, m_l\rangle = \sqrt{l(l+1) - m_l(m_l \pm 1)} |l, m_l \pm 1\rangle. \quad (\text{C.33})$$

Certain states have no orbital angular momentum. Singlet states ( $S = 0$ ) are a good example, and this arises because they have real wave functions, as can be proved straightforwardly. In the absence of a magnetic field the Hamiltonian is real so that if  $\psi$  is an eigenfunction of the Hamiltonian  $\hat{\mathcal{H}}$  with energy  $E$ , then so is  $\psi^*$ . ( $\hat{\mathcal{H}}\psi = E\psi$  so  $\hat{\mathcal{H}}\psi^* = E\psi^*$ .) But since the state is by definition a singlet,  $\psi = \psi^*$  and  $\psi$  is then real. The operators for all components of  $\hat{\mathbf{L}}$  contain  $i$  so that the expectation values  $\langle \psi | \hat{L}_\alpha | \psi \rangle$  for  $\alpha = x, y, z$  are all purely imaginary if  $\psi$  is real. But expectation values are measurable quantities and therefore must be pure real; hence they must all be zero and the state has no orbital angular momentum.

## C.5 The hydrogen atom

Most of the atoms considered in this book are not hydrogen, but the Schrödinger equation can only be solved exactly for the hydrogen atom. In this section, some results concerning the hydrogen atom are tabulated for easy reference.

The Schrödinger equation for a spherically symmetrical potential can be solved by separation of variables and the eigenfunction  $\Psi$  can be written as a

Table C.2 Radial wave functions.

| $n$ | $l$ | $L_{n+l}^{2l+1}(x)$           | $R_{nl}(\rho)$                                              |
|-----|-----|-------------------------------|-------------------------------------------------------------|
| 1   | 0   | $L_1^1(x) = -1$               | $2(Z/a_0)^{3/2}e^{-\rho/2}$                                 |
| 2   | 0   | $L_2^1(x) = 2x - 4$           | $(1/2\sqrt{2})(Z/a_0)^{3/2}(2 - \rho)e^{-\rho/2}$           |
| 2   | 1   | $L_3^3(x) = -6$               | $(1/2\sqrt{6})(Z/a_0)^{3/2}\rho e^{-\rho/2}$                |
| 3   | 0   | $L_3^1(x) = -3x^2 + 18x - 18$ | $(1/9\sqrt{3})(Z/a_0)^{3/2}(6 - 6\rho + \rho^2)e^{-\rho/2}$ |
| 3   | 1   | $L_4^3(x) = 24x - 96$         | $(1/9\sqrt{6})(Z/a_0)^{3/2}(4 - \rho)\rho e^{-\rho/2}$      |
| 3   | 2   | $L_5^5(x) = -120$             | $(1/9\sqrt{30})(Z/a_0)^{3/2}\rho^2 e^{-\rho/2}$             |

product of radial and angular parts:

$$\Psi = R(r)\Theta(\theta)\Phi(\phi) \quad (\text{C.34})$$

The equation for  $\Phi(\phi)$  is

$$\frac{d^2\Phi}{d\phi^2} + m_l^2\Phi = 0 \quad (\text{C.35})$$

which can be readily solved to yield solutions of the form

$$\Phi = Ae^{im_l\phi}. \quad (\text{C.36})$$

Solutions to the radial part of the wave function can be found and are of the form

$$R_{nl}(r) = -2 \left( \frac{Z}{na_0} \right)^{\frac{3}{2}} \left[ \frac{(n-l-1)!}{n[(n+l)!]^3} \right]^{\frac{1}{2}} \rho^l e^{-\rho/2} L_{n+l}^{2l+1}(\rho) \quad (\text{C.37})$$

where  $n$  is the principal quantum number,

$$\rho = \frac{2Zr}{na_0}, \quad (\text{C.38})$$

and the associated Laguerre polynomials are given by

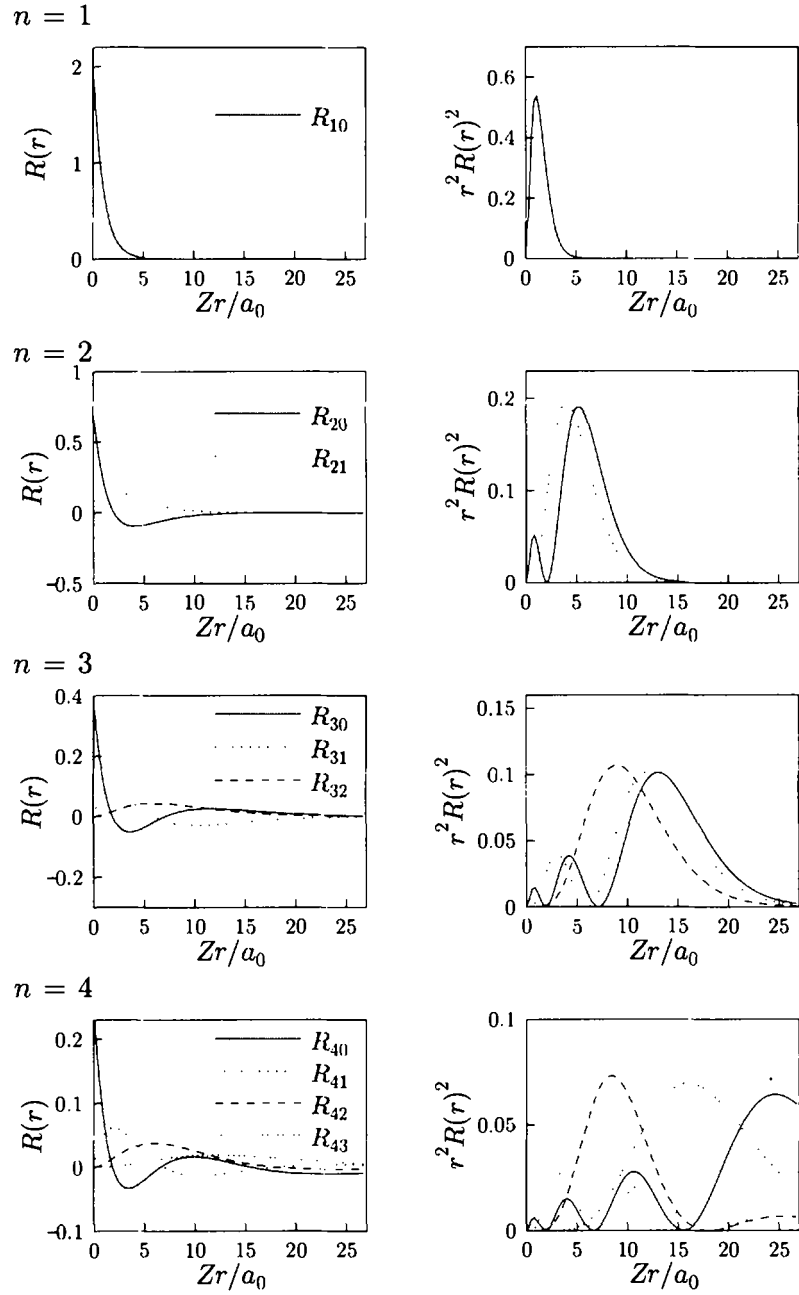
$$L_{\alpha}^{\beta}(x) = \frac{d^{\beta}}{dx^{\beta}} \left( e^x \frac{d^{\alpha}}{dx^{\alpha}} [e^{-x} x^{\alpha}] \right) = \sum_{j=0}^{\alpha-\beta} \frac{\alpha!^2 (-1)^{j+\beta}}{(j+\beta)! j! (\alpha-\beta-j)!} x^j. \quad (\text{C.39})$$

For atomic states the quantum numbers are in the range  $n \geq 1$ ,  $0 \leq l \leq n-1$  and  $-l \leq m_l \leq l$ . Table C.2 lists some radial wave functions for  $n = 1, 2, 3$ . Some radial wave functions  $R(r)$  for the hydrogen atom are plotted in Fig. C.2. Also shown is the radial probability density  $r^2 R(r)^2$ .

## C.6 The g-factor

The energy of the electron in a magnetic field  $B$  is

$$E = g\mu_B m_s B \quad (\text{C.40})$$



**Fig. C.2** The radial wave functions  $R(r)$  (left) and the corresponding probability density  $r^2 R(r)^2$  (right) for  $n = 1, 2, 3$  and 4.

where  $g$  is known as the g-factor. The energy levels therefore split by an amount  $g\mu_B B$ . A natural consequence of Dirac's theory of the electron (outside the scope of this book) is that  $g$  is precisely equal to 2. Actually the g-factor is not quite 2 but takes the value

$$g = 2\left(1 + \frac{\alpha}{2\pi} + \dots\right) = 2.0023 \dots \quad (\text{C.41})$$

where  $\alpha$  is the **fine structure constant**, a dimensionless quantity given by

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} = \frac{1}{137.04}. \quad (\text{C.42})$$

This theoretical value obtained from quantum electrodynamics (QED) agrees with experiment to an astonishing degree of precision. The discrepancy from  $g = 2$  can be explained as follows: electromagnetic interactions are due to the action of virtual photons. An electron can in fact emit a virtual photon, only to later reabsorb it. If you measure the magnetic moment of the electron during the life of one of these virtual photons, you will actually be measuring the magnetic moment of the electron–photon pair which will include an extra orbital component. You may also measure the system during a time in which more than one virtual photon has been created, though this sort of process becomes less probable as the number of virtual photons increases. On average then, the system has a magnetic moment slightly higher than what you might first expect, and this accounts for the non-zero value of  $g - 2$ . The expression is a power series in  $\alpha$  with each successive term reflecting the contributions of progressively more convoluted creations and absorptions of virtual photons.

## C.7 d orbitals

The angular parts of the d wave functions ( $l = 2$ ), in the form most often considered in condensed matter physics, can be constructed as linear combinations of functions of the form  $Y_{2m}e^{im_i\phi}$  where the functions  $Y_{2m}$  are those listed in the  $l = 2$  row of Table C.1:

$$Y_{xy} = \frac{Y_{22} - Y_{2-2}}{i\sqrt{2}} \quad (\text{C.43})$$

$$Y_{x^2-y^2} = \frac{Y_{22} + Y_{2-2}}{\sqrt{2}} \quad (\text{C.44})$$

$$Y_{yz} = \frac{-Y_{21} - Y_{2-1}}{i\sqrt{2}} \quad (\text{C.45})$$

$$Y_{zx} = \frac{-Y_{21} + Y_{2-1}}{\sqrt{2}} \quad (\text{C.46})$$

$$Y_{z^2} = Y_{20} \quad (\text{C.47})$$

which yields the following results (with the prefactors included for normalization)

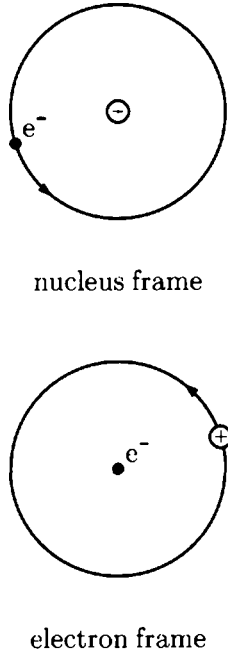
$$Y_{xy} = \sqrt{\frac{15}{16\pi}} \sin^2 \theta \sin 2\phi = \sqrt{\frac{15}{4\pi}} \frac{xy}{r^2} \quad (\text{C.48})$$

$$Y_{x^2-y^2} = \sqrt{\frac{15}{16\pi}} \sin^2 \theta \cos 2\phi = \sqrt{\frac{15}{16\pi}} \frac{x^2 - y^2}{r^2} \quad (\text{C.49})$$

$$Y_{yz} = \sqrt{\frac{15}{4\pi}} \sin \theta \cos \theta \sin \phi = \sqrt{\frac{15}{4\pi}} \frac{yz}{r^2} \quad (\text{C.50})$$

$$Y_{zx} = \sqrt{\frac{15}{4\pi}} \sin \theta \cos \theta \cos \phi = \sqrt{\frac{15}{4\pi}} \frac{zx}{r^2} \quad (\text{C.51})$$

$$Y_{z^2} = \sqrt{\frac{15}{16\pi}} (3 \cos^2 \theta - 1) = \sqrt{\frac{15}{16\pi}} \frac{2z^2 - x^2 - y^2}{r^2}. \quad (\text{C.52})$$



**Fig. C.3** The intrinsic spin-orbit interaction. In the frame of the nucleus the electron orbits the nucleus, but in an inertial frame comoving with the electron it is the nucleus which orbits.

<sup>2</sup>There are an infinite set of Lorentz transformations corresponding to the various instantaneous rest frames of the electron around its orbit. The Lorentz transformations do not commute when the direction of velocity changes and this is something our derivation has ignored. The net Lorentz transformation contains a rotation and it turns out that this Thomas precession yields a factor of precisely one-half.

L. H. Thomas (1903–1992)

These orbitals, together with the s and p orbitals, are shown in Fig. 3.1.

## C.8 The spin-orbit interaction

The spin-orbit interaction in an atom arises as follows. Consider an electron orbiting an atom. This is depicted at the top of Fig. C.3 in the nucleus rest frame. In the lower diagram, the atom is shown in an inertial frame comoving with the electron in which the nucleus appears to be orbiting the electron. The orbiting nucleus constitutes a current which gives rise to a magnetic field at the origin equal to

$$\mathbf{B} = \frac{\mathbf{E} \times \mathbf{v}}{c^2}, \quad (\text{C.53})$$

where

$$\mathbf{E} = -\nabla V(r) = -\frac{\mathbf{r}}{r} \frac{dV(r)}{dr} \quad (\text{C.54})$$

is the electric field at the electron due to the nucleus and  $V(r)$  is the corresponding potential energy. Equation C.53 comes from the transformation of electric and magnetic fields in special relativity. This magnetic field interacts with the spin of the electron to give a term in the Hamiltonian

$$H_{so} = -\frac{1}{2} \mathbf{m} \cdot \mathbf{B} \quad (\text{C.55})$$

$$= \frac{e\hbar^2}{2m_e c^2 r} \frac{dV(r)}{dr} \mathbf{S} \cdot \mathbf{L} \quad (\text{C.56})$$

where the orbital angular momentum is given by  $\hbar \mathbf{L} = m_e \mathbf{r} \times \mathbf{v}$  and the magnetic moment  $\mathbf{m} = (ge\hbar/2m)\mathbf{S}$ , and where the factor of  $\frac{1}{2}$  in eqn C.55 is the relativistic Thomas factor. This result can be obtained extremely elegantly with the relativistic correction automatically included using the Dirac equation.<sup>2</sup> This effect is known as the intrinsic spin-orbit interaction and is an interaction between the spin and the orbital part of an electron's wave function in an atom. For the Coulomb field in a hydrogen-like atom

$$\frac{1}{r} \frac{dV(r)}{dr} = \frac{Ze}{4\pi\epsilon_0 r^3} \quad (\text{C.57})$$

and for electronic states with quantum numbers  $l$  and  $n$  one has

$$\langle r^{-3} \rangle = \frac{Z^3}{a_0^3 n^3 l(l + \frac{1}{2})(l + 1)} \quad (\text{C.58})$$

so that the spin-orbit splitting is given by

$$\frac{Z^4 e^2 \hbar^2 \langle \mathbf{S} \cdot \mathbf{L} \rangle}{4\pi\epsilon_0 a_0^3 n^3 l(l + \frac{1}{2})(l + 1)}. \quad (\text{C.59})$$

## C.9 Landé g-factor

The Landé g-factor  $g_J$  is given by

$$g_J = \frac{3}{2} + \frac{S(S + 1) - L(L + 1)}{2J(J + 1)} \quad (\text{C.60})$$

The expression for  $g_J$  in eqn C.60 is obtained by starting with an expression for the magnetic moment operator:

$$\hat{\mu} = \mu_B(g_L \hat{\mathbf{L}} + g_S \hat{\mathbf{S}}). \quad (\text{C.61})$$

In this expression  $g_L = 1$  and  $g_S = 2$  are the g-factors for orbital and spin angular momentum respectively. Now  $\mathbf{S}$  and  $\mathbf{L}$  may not be good quantum numbers in many atoms, but  $\mathbf{J}$  is. Therefore the component of the magnetic moment which is parallel to  $\mathbf{J}$  will be a conserved quantity but the component perpendicular to  $\mathbf{J}$  will not be. We therefore write

$$\hat{\mu} = g_J \mu_B \hat{\mathbf{J}}, \quad (\text{C.62})$$

where  $g_J$  is a constant to be determined. Thus eqn C.62 implies that  $g_J$  is the projection of  $\hat{\mathbf{L}} + 2\hat{\mathbf{S}}$  on to  $\hat{\mathbf{J}}$ . Multiplying both sides of eqn C.61 by  $\hat{\mathbf{J}}$  yields

$$\hat{\mu} \cdot \mathbf{J} = \mu_B(g_L \hat{\mathbf{L}} \cdot \hat{\mathbf{J}} + g_S \hat{\mathbf{S}} \cdot \hat{\mathbf{J}}). \quad (\text{C.63})$$

Multiplying both sides of eqn C.62 by  $\hat{\mathbf{J}}$  yields<sup>3</sup>

$$\hat{\mu} \cdot \mathbf{J} = g_J \mu_B \hat{\mathbf{J}}^2 = g_J \mu_B J(J+1). \quad (\text{C.64})$$

<sup>3</sup>In this, and subsequent steps, we use the fact that the eigenvalue of  $\mathbf{S}^2$  is  $S(S+1)$ , of  $\mathbf{L}^2$  is  $L(L+1)$  and of  $\mathbf{J}^2$  is  $J(J+1)$ .

Also,  $\mathbf{L}^2 = (\mathbf{J} - \mathbf{S})^2 = \mathbf{J}^2 + \mathbf{S}^2 - 2\mathbf{S} \cdot \mathbf{J}$  so that

$$\mathbf{S} \cdot \mathbf{J} = \frac{1}{2}(\mathbf{J}^2 - \mathbf{L}^2 + \mathbf{S}^2) \quad (\text{C.65})$$

and  $\mathbf{S}^2 = (\mathbf{J} - \mathbf{L})^2 = \mathbf{J}^2 + \mathbf{L}^2 - 2\mathbf{L} \cdot \mathbf{J}$  so that

$$\mathbf{L} \cdot \mathbf{J} = \frac{1}{2}(\mathbf{J}^2 + \mathbf{L}^2 - \mathbf{S}^2). \quad (\text{C.66})$$

Equating eqns C.63 and C.64, and inserting the results from eqns C.65 and C.66 yields

$$\begin{aligned} g_J = g_L & \left( \frac{J(J+1) + L(L+1) - S(S+1)}{2J(J+1)} \right) \\ & + g_S \left( \frac{J(J+1) - L(L+1) + S(S+1)}{2J(J+1)} \right) \end{aligned} \quad (\text{C.67})$$

which reduces to eqn C.60 if  $g_L = 1$  and  $g_S = 2$ .

## C.10 Perturbation theory

Consider a Hamiltonian  $\hat{\mathcal{H}}_0$  which has known eigenfunctions  $|\phi_i\rangle$  with known eigenvalues  $E_i$ , so that

$$\hat{\mathcal{H}}_0 |\phi_i\rangle = E_i |\phi_i\rangle. \quad (\text{C.68})$$

The eigenfunctions  $|\phi_i\rangle$  are all orthogonal so that

$$\langle \phi_i | \phi_j \rangle = \int d\tau \phi_i^* \phi_j = \delta_{ij} \quad (\text{C.69})$$

where  $d\tau$  is a volume element. Now suppose that a perturbation  $\hat{V}$  is added to  $\hat{\mathcal{H}}_0$  so that the new Hamiltonian is  $\hat{\mathcal{H}}$  where

$$\hat{\mathcal{H}} = \hat{\mathcal{H}}_0 + \hat{V}. \quad (\text{C.70})$$

The system starts in a state  $|\phi_k\rangle$  with energy  $E_k$  before the addition of the perturbation. The new eigenfunction of the system will be  $|\psi\rangle$  and the eigenvalue  $E$  where

$$\hat{\mathcal{H}}|\psi\rangle = E|\psi\rangle, \quad (\text{C.71})$$

but both  $|\psi\rangle$  and  $E$  are unknown so far.  $|\psi\rangle$  can be expanded in terms of the old eigenfunctions so that

$$|\psi\rangle = \sum_j a_j |\phi_j\rangle. \quad (\text{C.72})$$

Inserting this into eqn C.71, premultiplying by  $\langle\phi_i|$  and integrating over the appropriate volume gives

$$\sum_j a_j \int d\tau \phi_i^* (\hat{\mathcal{H}}_0 + \hat{V}) \phi_j = E \sum_j a_j \int d\tau \phi_i^* \phi_j, \quad (\text{C.73})$$

so that using the orthogonality of the original eigenfunctions (eqn C.69) yields

$$\sum_j V_{ij} a_j = (E - E_i) a_i, \quad (\text{C.74})$$

which is known as the **secular equation**, where

$$V_{ij} = \langle\phi_i|\hat{V}|\phi_j\rangle = \int d\tau \phi_i^* \hat{V} \phi_j \quad (\text{C.75})$$

From now on, we shall assume the non-degenerate case, i.e. we will assume that none of the states are degenerate.

is the **matrix element** of the perturbation.

If the perturbation is small,  $|\psi\rangle$  will be very close to the starting state  $|\phi_k\rangle$  so that  $a_k \approx 1$  and  $|a_i| \ll 1$  for  $i \neq k$ . Since the wave function has changed only marginally, one might imagine that the new energy is given by

$$E \approx E_k, \quad (\text{C.76})$$

in other words that it remains completely unchanged. We can improve on this zeroth order approximation by using the results  $a_k \approx 1$  and  $a_i \ll 1$  for  $i \neq k$ . Substituting these into the secular equation can give some approximate expressions for the energy and for  $a_i$ . For  $i = k$ , the sum in the secular equation is dominated by the term for which  $j = k$  giving

$$E \approx E_k + V_{kk}, \quad (\text{C.77})$$

the shift of the energy of the state is given simply by  $V_{kk}$ . This result, which contains the ‘first-order perturbation theory’ correction to the energy, is used very often in the text. The secular equation can also be evaluated for the case  $i \neq k$ , yielding

$$a_i = \frac{V_{ik}}{E_k - E_i} \quad (\text{C.78})$$

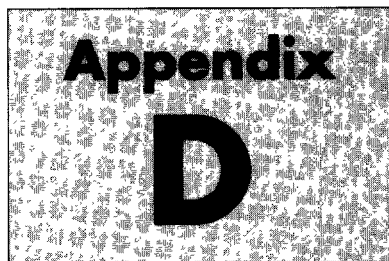
(where  $E$  has been replaced by  $E_k$ ). Putting eqn C.78 back into eqn C.74 yields the next level of approximation for the energy, namely

$$E \approx E_k + V_{kk} + \sum_{i \neq k} \frac{|V_{ik}|^2}{E_k - E_i} \quad (\text{C.79})$$

which is the ‘second-order perturbation theory’ result. This process can be continued and will generate a power series expression for  $E$ , with each successive term in the series containing a higher power of the matrix elements of the perturbation.

## Further reading

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# Energy in magnetism and demagnetizing fields

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## D.1 Energy

Calculating the energy of a magnetized medium in a magnetic field is a surprisingly subtle business. Various expressions can be obtained but they can refer to different things. In magnetism it depends on whether one is talking about the energy of the magnetic moment alone, or the magnetic moment plus whatever it is that is providing the magnetic field, and exactly how this energy is partitioned up.

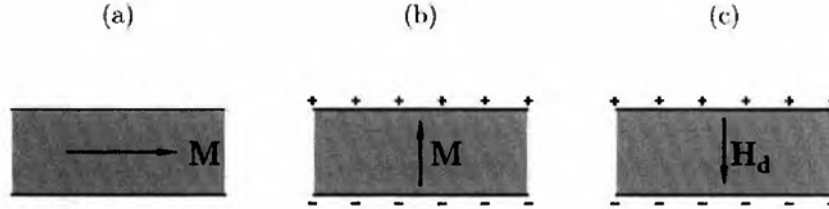
For example, if you place a screwdriver close to the pole pieces of a large magnet, the screwdriver is strongly attracted into the region of maximum field. Replacing a bit of the air in the gap between the pole pieces with the magnetized screwdriver therefore seems to be an energetically favourable thing, suggesting that the presence of the magnetizable medium (the screwdriver) is lowering the energy. However, it takes more energy to establish a current in a magnet which contains an iron core than in one without an iron core, suggesting that the presence of the magnetizable medium (in this case the iron core) is increasing the energy. So does the energy decrease or increase? It depends on which energy you are considering. In the first example we are only measuring the energy saved by the force we feel on our arm holding the screwdriver, and we are ignoring the extra work needed to be done by the magnet power supply in maintaining the current at the level it had before we started waving the screwdriver around. This emphasizes that great care is needed in treating the energetics of magnetic materials (for a detailed treatment, see the article by Heine in the further reading).

In Appendix B, we found that  $\delta W = -\boldsymbol{\mu} \cdot \delta \mathbf{B}$  is the appropriate free energy to consider for magnetized media. This expression will be useful in formulating magnetism in statistical mechanics (see Appendix E). In this appendix, we concentrate on the way in which the presence of a ferromagnetic material can change both  $\mathbf{H}$  and  $\mathbf{B}$  because of the introduction of demagnetizing fields which themselves cost energy.

## D.2 Demagnetizing factors

When the magnetization  $\mathbf{M}$  inside a ferromagnetic body meets the surface, it has to suddenly stop. Hence there is a divergence of  $\mathbf{M}$ . Using the equation

$$\nabla \cdot \mathbf{H} = -\nabla \cdot \mathbf{M}, \quad (\text{D.1})$$



**Fig. D.1** Demagnetization in an infinite flat plate, viewed in cross-section. (a) If the magnetization lies in the plane of the plate, no magnetic poles are created on the surface of the plate (except for tiny ones at the ends). (b) If the magnetization is perpendicular to the plane of the plate, as shown, there is a negative (positive) divergence of  $\mathbf{M}$  on the top (bottom) surface which produces a positive (negative) divergence of  $\mathbf{H}$ . This results in positive (negative) magnetic poles on the top (bottom) surface as shown. (c) This results in a demagnetization field which runs from positive magnetic poles to negative magnetic poles.

(which follows from  $\nabla \cdot \mathbf{B} = 0$ , see appendix B) we find that there is an equal and opposite divergence of  $\mathbf{H}$ . The situation is as if magnetic monopoles have been left on the surface of the ferromagnet, and these monopoles act as sources of  $\mathbf{H}$ . The resulting  $\mathbf{H}$  field is known as a **demagnetizing field**. This situation is illustrated for a simple case in Figure D.1 which describes the magnetization in an infinite flat plate or thin film of ferromagnetic material. If the magnetization lies in the plane of the plate, the only divergence of  $\mathbf{M}$  is at the ends which, we suppose, are an infinite distance away. Therefore there is no demagnetizing field. If the magnetization is perpendicular to the plane of the plate, magnetic poles are created on top and bottom surfaces and give rise to a demagnetizing field  $\mathbf{H}_d = -\mathbf{M}$  inside the plate.

The demagnetizing field can be an extremely complicated function of position for a ferromagnet of arbitrary shape. However, it takes a relatively simple form in the case of an ellipsoidal ferromagnet. In this case it is uniform inside the ferromagnet with a value of  $\mathbf{H}_d$  equal to

$$\mathbf{H}_d = -\mathbf{N}\mathbf{M} \quad (\text{D.2})$$

where  $\mathbf{N}$  is the demagnetizing tensor. Thus in general we can write

$$(H_d)_i = -\sum_j N_{ij} M_j. \quad (\text{D.3})$$

If  $\mathbf{M}$  is along one of the principal axes of the ellipse,  $N_{ij}$  can be diagonalized so that

$$\mathbf{N} = \begin{pmatrix} N_x & 0 & 0 \\ 0 & N_y & 0 \\ 0 & 0 & N_z \end{pmatrix}. \quad (\text{D.4})$$

The demagnetizing tensor satisfies

$$\text{Tr}\mathbf{N} = N_x + N_y + N_z = 1. \quad (\text{D.5})$$

### Example D.1

We now examine some special cases.

- (1) For a sphere,  $N_x = N_y = N_z = \frac{1}{3}$ , so that

$$\mathbf{H}_d = -\frac{\mathbf{M}}{3} \quad (\text{D.6})$$

- (2) For a very long cylindrical rod parallel to  $z$ ,  $N_x = N_y = \frac{1}{2}$ ,  $N_z = 0$ . This is because if the magnetization lies exactly along the rod, the magnetic poles created will be at either end of the rod which can be assumed to be too far away to matter.
- (3) For a flat plate perpendicular to  $z$ ,  $N_x = N_y = 0$ ,  $N_z = 1$ . This is the case considered above.

### D.3 A ferromagnet of arbitrary shape

In this section we will show how to compute the energy due to the demagnetizing energy for a ferromagnet of arbitrary shape in an applied magnetic field. For this problem, the magnetic fields  $\mathbf{H}(\mathbf{r})$  and  $\mathbf{B}(\mathbf{r})$  can be broken up into two components:

$$\mathbf{H}(\mathbf{r}) = \mathbf{H}_a(\mathbf{r}) + \mathbf{H}_d(\mathbf{r}) \quad (\text{D.7})$$

$$\mathbf{B}(\mathbf{r}) = \mathbf{B}_a(\mathbf{r}) + \mathbf{B}_d(\mathbf{r}), \quad (\text{D.8})$$

where the applied field is  $\mathbf{H}_a(\mathbf{r}) = \mathbf{B}_a(\mathbf{r})/\mu_0$  and the demagnetizing field is  $\mathbf{H}_d(\mathbf{r})$  and we write  $\mathbf{B}_d(\mathbf{r})$  as a sum of the flux density due to the demagnetizing field  $\mu_0\mathbf{H}_d(\mathbf{r})$  and  $\mu_0\mathbf{M}(\mathbf{r})$  inside the material, i.e.  $\mathbf{B}_d(\mathbf{r}) = \mu_0(\mathbf{H}_d(\mathbf{r}) + \mathbf{M}(\mathbf{r}))$ . The demagnetizing field arises from the magnetic poles which are produced on the surface of the magnetic body wherever  $\nabla \cdot \mathbf{M}(\mathbf{r}) \neq 0$ .

We now state an important result. The demagnetizing fields  $\mathbf{B}_d(\mathbf{r})$  and  $\mathbf{H}_d(\mathbf{r})$  satisfy

$$\int_{\text{all space}} \mathbf{B}_d \cdot \mathbf{H}_d \, d\tau = 0. \quad (\text{D.9})$$

The proof is as follows: since  $\nabla \times \mathbf{H}_d = 0$ , the demagnetizing field  $\mathbf{H}_d$  can be written in terms of the gradient of a scalar function  $\phi$  (e.g.  $\mathbf{H}_d = -\nabla\phi$ ). Now

$$\nabla \cdot (\phi \mathbf{B}_d) \equiv \phi \nabla \cdot \mathbf{B}_d + (\nabla\phi) \cdot \mathbf{B}_d = (\nabla\phi) \cdot \mathbf{B}_d \quad (\text{D.10})$$

where the last equality follows from  $\nabla \cdot \mathbf{B}_d = 0$ . Then

$$\int_{\text{all space}} \mathbf{B}_d \cdot \mathbf{H}_d \, d\tau = - \int_{\text{all space}} \nabla \cdot (\phi \mathbf{B}_d) \, d\tau = \lim_{R \rightarrow \infty} \int_{\text{surface}} \phi \mathbf{B}_d \cdot d\mathbf{S} = 0, \quad (\text{D.11})$$

where the surface integral is taken over a sphere whose radius  $R$  tends to infinity. The integral is zero if the ferromagnetic body is of finite extent, so that as  $R \rightarrow \infty$ ,  $\mathbf{B}_d \sim R^{-2}$  and  $\phi \sim R^{-1}$  at worst.

If the applied field is not present, then the energy  $E$  of the demagnetizing field is simply the integral over all space of the energy density  $\frac{1}{2}\mu_0\mathbf{H}_d^2$ . This

can be converted to an integral inside the volume  $V$  of the ferromagnetic body by using eqn D.9, so that the energy can be expressed as

$$\begin{aligned} E &= \frac{\mu_0}{2} \int_{\text{all space}} \mathbf{H}_d^2 d\tau = -\frac{1}{2} \int_{\text{all space}} (\mathbf{B}_d - \mu_0 \mathbf{M}) \cdot \mathbf{H}_d d\tau \\ &= -\frac{\mu_0}{2} \int_{\text{all space}} \mathbf{M} \cdot \mathbf{H}_d d\tau \\ &= -\frac{\mu_0}{2} \int_V \mathbf{M} \cdot \mathbf{H}_d d\tau, \end{aligned} \quad (\text{D.12})$$

where the last line follows from the fact that  $\mathbf{M} = 0$  outside the body. It should be noted that even though this final integral is only over the body, it expresses the energy of all stray fields, including those *outside* the body.

If there is an applied field, then the energy of the ferromagnetic body may be written as the difference between the energy density in the total field  $\frac{1}{2}\mu_0 \mathbf{H}^2$  and the energy density in the applied field  $\frac{1}{2}\mu_0 \mathbf{H}_a^2$ . This removes the infinite energy contribution from a uniform applied field integrated over all space. Thus the energy can be usefully written as

$$E = \frac{\mu_0}{2} \int_{\text{all space}} (\mathbf{H}^2 - \mathbf{H}_a^2) d\tau. \quad (\text{D.13})$$

Using the fact that  $\nabla \times \mathbf{H}_a = 0$ , we have

$$\int_{\text{all space}} \mathbf{H}_a \cdot \mathbf{B}_d d\tau = 0. \quad (\text{D.14})$$

Hence, by also using the following equations:

$$\mathbf{H}^2 = \mathbf{H}_a^2 + 2\mathbf{H}_d \cdot \mathbf{H}_a + \mathbf{H}_d^2, \quad (\text{D.15})$$

$$\mathbf{H}_d \cdot \mathbf{H}_a = \mathbf{H}_a \cdot \left( \frac{\mathbf{B}_d}{\mu_0} - \mathbf{M} \right), \quad (\text{D.16})$$

and by also using eqn D.9, the energy  $E$  can be written in the familiar form

$$E = -\mu_0 \int_V \mathbf{M} \cdot \mathbf{H}_a d\tau - \frac{\mu_0}{2} \int_V \mathbf{M} \cdot \mathbf{H}_d d\tau, \quad (\text{D.17})$$

the sum of a Zeeman term and a demagnetizing term, with each integral being taken only over the volume  $V$  of the ferromagnetic body. The energy in eqn D.17 nevertheless implicitly includes the energy associated with the demagnetizing fields outside the body.

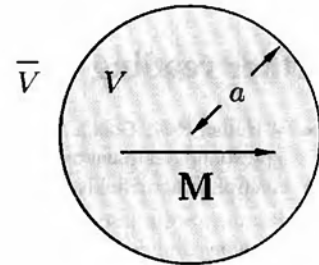
### Example D.2

A ferromagnetic sphere of radius  $a$  with uniform magnetization  $\mathbf{M}$  is used here as a detailed example of the ideas leading to eqn D.12. It is sketched in Fig. D.2. The demagnetizing fields are given by

$$\mathbf{H}_d = -\frac{\mathbf{M}}{3} \quad (\text{D.18})$$

$$\mathbf{B}_d = \frac{2\mu_0 \mathbf{M}}{3}. \quad (\text{D.19})$$

The factor of one-half in the demagnetizing term can be understood in terms of an avoidance of double-counting. The Zeeman term is the interaction of the magnetization with the applied field. The demagnetizing term is a self-energy so the moments which cause the field interact with their own field. The factor of one-half ensures that you don't count this energy twice.



**Fig. D.2** A ferromagnetic sphere with uniform magnetization  $\mathbf{M}$ . The region inside the sphere of volume  $V_{\text{sphere}} = \frac{4}{3}\pi a^3$  is denoted  $V$ , the region outside the sphere is denoted  $\bar{V}$ .

so that inside the sphere (a space of volume  $V_{\text{sphere}} = \frac{4}{3}\pi a^3$ , denoted by  $V$ ) simple integration yields the following results:

$$-\frac{\mu_0}{2} \int_V \mathbf{M} \cdot \mathbf{H}_d d\tau = \frac{1}{6} \mu_0 M^2 V_{\text{sphere}} \quad (\text{D.20})$$

$$\frac{\mu_0}{2} \int_V \mathbf{H}_d^2 d\tau = \frac{1}{18} \mu_0 M^2 V_{\text{sphere}} \quad (\text{D.21})$$

$$\frac{1}{2} \int_V \mathbf{B}_d \cdot \mathbf{H}_d d\tau = -\frac{1}{9} \mu_0 M^2 V_{\text{sphere}} \quad (\text{D.22})$$

Outside the sphere (a space denoted by  $\bar{V}$ ),  $\mathbf{B}_d = \mu_0 \mathbf{H}_d$  (since  $\mathbf{M} = 0$  outside the sphere), and the demagnetizing fields result from the magnetic dipole moment of the sphere. These have radial and polar components given by

$$\begin{aligned} (H_d)_r &= 2Ma^3 \cos \theta / 3r^3 \\ (H_d)_\theta &= Ma^3 \sin \theta / 3r^3, \end{aligned} \quad (\text{D.23})$$

so that integration over this region outside the sphere gives:

$$\frac{\mu_0}{2} \int_{\bar{V}} \mathbf{H}_d^2 d\tau = \frac{1}{2} \int_{\bar{V}} \mathbf{B}_d \cdot \mathbf{H}_d d\tau = \frac{1}{2\mu_0} \int_{\bar{V}} \mathbf{B}_d^2 d\tau = \frac{1}{9} \mu_0 M^2 V_{\text{sphere}}. \quad (\text{D.24})$$

Then integration over all space gives

$$\begin{aligned} -\frac{\mu_0}{2} \int_{\text{all space}} \mathbf{M} \cdot \mathbf{H}_d d\tau &= \frac{\mu_0}{2} \int_{\text{all space}} \mathbf{H}_d^2 d\tau \\ &= \frac{1}{6} \mu_0 M^2 V_{\text{sphere}} \end{aligned} \quad (\text{D.25})$$

in agreement with eqn D.12. By combining eqns D.22 and D.24 one has

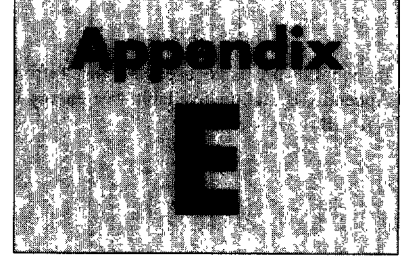
$$\frac{1}{2} \int_{\text{all space}} \mathbf{B}_d \cdot \mathbf{H}_d d\tau = 0 \quad (\text{D.26})$$

in agreement with eqn D.9.

## Further reading

- V. Heine, *Proc. Camb. Phil. Soc.* **52**, 546 (1956) contains a profound treatment of the thermodynamics of bodies in electromagnetic fields.
- Treatments are also found in C. J. Adkins, *Equilibrium thermodynamics*, CUP 1983, and J. R. Waldram, *The theory of thermodynamics*, CUP 1985.
- Magnetostatic energy and demagnetizing factors are covered in depth in A. Aharoni, *Introduction to the theory of ferromagnetism*, OUP 1996.
- I. Brevik, *J. Phys.: Condensed Matter* **7**, 8065 (1995) contains a helpful analysis of the problem of calculating the energy of a ferromagnetic body immersed in a ferromagnetic liquid.

# Statistical mechanics



This appendix briefly sketches some results in statistical mechanics which will be useful in the main text.

## E.1 The partition function and thermodynamic functions

The probability  $p_i$  of occupying a state  $i$  with energy  $E_i$  at temperature  $T$  is

$$p_i = \frac{e^{-E_i/k_B T}}{Z} \quad (\text{E.1})$$

where  $Z$  is the single-particle partition function which is used to normalize the probability such that

$$\sum_i p_i = 1. \quad (\text{E.2})$$

This implies that  $Z$  is given by

$$Z = \sum_i e^{-E_i/k_B T}. \quad (\text{E.3})$$

Defining  $\beta = 1/k_B T$ ,  $\partial Z/\partial \beta$  can be evaluated, giving

$$\frac{\partial Z}{\partial \beta} = - \sum_i E_i e^{-\beta E_i}. \quad (\text{E.4})$$

Now since we have found the probability distribution, we can use it to calculate expected values of quantities. The expected value of the energy  $\langle E \rangle$  is given by

$$\langle E \rangle = \sum_i E_i p_i = \frac{\sum_i E_i e^{-\beta E_i}}{Z} = - \frac{1}{Z} \frac{\partial Z}{\partial \beta} = - \frac{\partial \ln Z}{\partial \beta}. \quad (\text{E.5})$$

The entropy per particle is given by

$$S = -k_B \sum_i p_i \ln p_i \quad (\text{E.6})$$

and hence

$$S = -k_B \sum_i \frac{e^{-\beta E_i}}{Z} [-\beta E_i - \ln Z] = k_B \ln Z + \frac{\langle E \rangle}{T}. \quad (\text{E.7})$$

E.1 The partition function and thermodynamic functions

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E.2 The equipartition theorem

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In eqn E.8 we have used the result from Appendix B that the available free energy is  $-\mu \cdot \delta \mathbf{B}$ .

For  $N$  independent distinguishable particles, the appropriate partition function becomes  $Z^N$  and so the total energy is  $E = N\langle E \rangle$ . The first law of thermodynamics can be expressed as

$$dE = TdS - pdV - Md\mathbf{B}. \quad (\text{E.8})$$

The **Helmholtz free energy**  $F$ , also known as the **Helmholtz function**, is defined by

$$F = E - TS, \quad (\text{E.9})$$

and is related to the partition function  $Z$  by

$$F = -Nk_B T \ln Z. \quad (\text{E.10})$$

Using eqns E.8 and E.9,  $dF$  can be written

$$dF = -SdT - pdV - Md\mathbf{B} \quad (\text{E.11})$$

and so

$$M = -\left(\frac{\partial F}{\partial \mathbf{B}}\right)_{T,V} = Nk_B T \left(\frac{\partial \ln Z}{\partial \mathbf{B}}\right)_{T,V}. \quad (\text{E.12})$$

The entropy is given by

$$M = -\left(\frac{\partial F}{\partial T}\right)_{V,B}. \quad (\text{E.13})$$

## E.2 The equipartition theorem

Very often the energy of a system can be written as a sum of quadratic terms. For example, a mass  $m$  on a spring of spring constant  $K$  has an energy  $E = \frac{1}{2}mv^2 + \frac{1}{2}Kx^2$ , which contains two such quadratic terms. At low temperatures, where quantum effects dominate, we expect harmonic oscillator solutions and a ladder of energy levels. But at high temperatures, where  $k_B T$  is much larger than the energy scale set by the spacing between the levels, the system can be considered classically.

The **equipartition theorem** states that in the classical limit, when  $k_B T$  is much larger than the energy spacing between quantum levels, and the energy contains a number of independent quadratic terms, the expected value of the energy is equal to  $\frac{1}{2}k_B T$  multiplied by the number of independent quadratic terms in the energy.

The proof is straightforward: if  $E(x_1, x_2, \dots, x_n) = \sum_{i=1}^n \alpha_i x_i^2$  where  $\alpha_i$  are constants, then

$$\langle E \rangle = \frac{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} dx_1 dx_2 \dots dx_n e^{-\beta E(x_1, x_2, \dots, x_n)} E}{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} dx_1 dx_2 \dots dx_n e^{-\beta E(x_1, x_2, \dots, x_n)}}. \quad (\text{E.14})$$

This looks formidable but the integrals for each variable can be easily

separated. Hence

$$\begin{aligned}
 \langle E \rangle &= \sum_{i=1}^n \alpha_i \left( \frac{\int_{-\infty}^{\infty} dx_i x_i^2 e^{-\beta \alpha_i x_i^2} \prod_{j \neq i} \int_{-\infty}^{\infty} dx_j e^{-\beta \alpha_j x_j^2}}{\int_{-\infty}^{\infty} dx_i e^{-\beta \alpha_i x_i^2} \prod_{j \neq i} \int_{-\infty}^{\infty} dx_j e^{-\beta \alpha_j x_j^2}} \right) \\
 &= \sum_{i=1}^n \left( \frac{\frac{\alpha_i}{2} \sqrt{\frac{\pi}{\alpha_i^3 \beta^3}}}{\sqrt{\frac{\pi}{\alpha_i \beta}}} \right) \\
 &= \sum_{i=1}^n \frac{1}{2\beta} \\
 &= \frac{n}{2} k_B T,
 \end{aligned} \tag{E.15}$$

where use has been made of the standard integrals

$$\int_{-\infty}^{\infty} dx e^{-\gamma x^2} = \sqrt{\frac{\pi}{\gamma}} \quad \text{and} \quad \int_{-\infty}^{\infty} dx x^2 e^{-\gamma x^2} = \frac{1}{2} \sqrt{\frac{\pi}{\gamma^3}}. \tag{E.16}$$

### Example E.1

Three examples of this result now follow:

- (1) For a mass  $m$  on a spring of spring constant  $K$ ,  $E = \frac{1}{2}mv^2 + \frac{1}{2}Kx^2$ , (two quadratic terms), so that  $\langle E \rangle = k_B T$ .
- (2) For a molecule of mass  $m$  in a gas,  $E = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2$ , (three quadratic terms), so that  $\langle E \rangle = \frac{3}{2}k_B T$ .
- (3) For a solid containing  $N$  atoms (and therefore  $3N$  springs, and hence the energy contains  $6N$  quadratic terms),  $\langle E \rangle = 3Nk_B T$ .

In addition to adding  $\frac{1}{2}k_B T$  to the energy, each quadratic degree of freedom adds  $\frac{1}{2}k_B$  to the heat capacity.

### Further reading

- F. Reif, *Fundamentals of statistical and thermal physics*, Mc-Graw-Hill, 1965.
- J. R. Waldram, *The theory of thermodynamics*, CUP 1985.
- P. M. Chaikin and T. C. Lubensky, *Principles of condensed matter physics*, CUP 1995.
- A. M. Glazer and J. S. Wark, *Statistical mechanics*, OUP 2001.

# Appendix F

## Answers and hints to selected problems

- (1.1)  $9.27 \times 10^{-24} \text{ Am}^2$ ;  $8.40 \text{ GHz}$ ;  $34.7 \mu\text{eV}$ ;  $8.40 \text{ GHz}$ . Here we are only considering *spin* angular momentum.

- (1.4) This is the commutator of a scalar operator ( $\hat{\mathbf{S}} \cdot \mathbf{X}$ ) and a vector operator ( $\hat{\mathbf{S}}$ ). Work out one component of this, and so evaluate

$$\begin{aligned} [\hat{\mathbf{S}} \cdot \mathbf{X}, \hat{\mathbf{S}} \cdot \mathbf{Y}] &= \frac{1}{4}((\sigma \cdot \mathbf{X})(\sigma \cdot \mathbf{Y}) - (\sigma \cdot \mathbf{Y})(\sigma \cdot \mathbf{X})) \\ &= \frac{i}{4}(\sigma \cdot (\mathbf{X} \times \mathbf{Y}) - \sigma \cdot (\mathbf{Y} \times \mathbf{X})) \\ &= \frac{i}{2}\sigma \cdot (\mathbf{X} \times \mathbf{Y}) \\ &= i\mathbf{S} \times \mathbf{X} \cdot \mathbf{Y}, \end{aligned} \quad (\text{F.1})$$

and the result follows.

- (1.5) Using eqn 1.59 one can show that

$$\begin{aligned} \hat{S}_z(\hat{S}_{\pm}|S, S_z\rangle) &= (\hat{S}_{\pm}\hat{S}_z \pm \hat{S}_{\pm})|S, S_z\rangle \\ &= (S_z \pm 1)(\hat{S}_{\pm}|S, S_z\rangle), \end{aligned} \quad (\text{F.2})$$

so that the z-component of the spin is raised or lowered. Equations 1.58 and 1.61 can be written as

$$\hat{S}_+\hat{S}_- - \hat{S}_-\hat{S}_+ = 2\hat{S}_z \quad (\text{F.3})$$

$$\begin{aligned} \hat{S}_+\hat{S}_- + \hat{S}_-\hat{S}_+ &= 2(\hat{S}_x^2 + \hat{S}_y^2) \\ &= \hat{\mathbf{S}}^2 - \hat{S}_z^2, \end{aligned} \quad (\text{F.4})$$

so that adding or subtracting leads to

$$\hat{S}_+\hat{S}_- = \hat{\mathbf{S}}^2 - \hat{S}_z^2 + \hat{S}_z \quad (\text{F.5})$$

$$\hat{S}_-\hat{S}_+ = \hat{\mathbf{S}}^2 - \hat{S}_z^2 - \hat{S}_z. \quad (\text{F.6})$$

Hence

$$\begin{aligned} \langle S, S_z | \hat{S}_-\hat{S}_+ | S, S_z \rangle &= S(S+1) - S_z^2 - S_z \\ &= S(S+1) - S_z(S_z+1) \end{aligned} \quad (\text{F.7})$$

and

$$\begin{aligned} \langle S, S_z | \hat{S}_+\hat{S}_- | S, S_z \rangle &= S(S+1) - S_z^2 + S_z \\ &= S(S+1) - S_z(S_z-1) \end{aligned} \quad (\text{F.8})$$

and the required normalization is therefore proved.

- (1.6) Without loss of generality, put  $\mathbf{B} = (0, 0, B)$  so that  $\mathbf{B} \times \mathbf{r} = B(-y, x, 0)$  and  $\frac{1}{2}\nabla \times (\mathbf{B} \times \mathbf{r}) = (0, 0, B)$ . Other choices are possible, e.g.  $\mathbf{A} = -By(1, 0, 0)$  or  $\mathbf{A} = Bx(0, 1, 0)$ . In fact,  $\mathbf{A} = B((\alpha-1)y, \alpha x, 0)$  will work where  $\alpha$  is a real number.

- (1.7) Hint: evaluate  $[\sigma \cdot (\mathbf{p} + e\mathbf{A})]^2/2m_e$ .

- (1.8) Without loss of generality, the new axis can be at an angle  $\theta$  to the z axis in the  $xz$  plane. Thus the operator is

$$\hat{S}_{\theta} = \cos\theta\hat{S}_z + \sin\theta\hat{S}_x. \quad (\text{F.9})$$

The eigenvector of this operator with eigenvalue 1 is found to be

$$(2(1 - \cos\theta))^{-1/2} \begin{pmatrix} \sin\theta \\ 1 - \cos\theta \end{pmatrix}, \quad (\text{F.10})$$

so that the required probability is

$$\frac{\sin^2\theta}{2(1 - \cos\theta)} = \cos^2\frac{\theta}{2}. \quad (\text{F.11})$$

- (1.10)  $H_d/H_a$ : (a)  $3 \times 10^{-5}$  (b)  $-8.8 \times 10^{-4}$

$$B_d/B_a$$
: (a)  $-6 \times 10^{-5}$  (b)  $1.76 \times 10^{-3}$

- (1.12) Hint:  $\sigma_m^2 = I$  so that

$$\sigma^n = \begin{cases} I & n \text{ even} \\ \sigma_m & n \text{ odd} \end{cases} \quad (\text{F.12})$$

so that

$$\begin{aligned} e^{i\alpha\sigma_m} &= I \left( 1 - \frac{\alpha^2}{2!} + \frac{\alpha^4}{4!} - \dots \right) \\ &\quad + i\sigma_m \left( \alpha - \frac{\alpha^3}{3!} + \frac{\alpha^5}{5!} - \dots \right) \\ &= I \cos\alpha + i\sigma_m \sin\alpha. \end{aligned} \quad (\text{F.13})$$

- (1.14) (a) Torque  $= I\ddot{\theta} = -p\mathcal{E} \sin\theta$  so that  $\ddot{\theta} + p\mathcal{E} \sin\theta/I = 0$  which leads to simple harmonic motion when  $\theta \ll 1$  with  $\omega^2 = p\mathcal{E}/I$ .  
(b) The Hamiltonian is  $\mathcal{H} = \hat{L}^2/2I - p\mathcal{E} \cos\theta$ . In this calculation, several commutators need to be evaluated. You should find that

$$[\hat{\theta}, \hat{L}] = i\hbar \quad (\text{F.14})$$

$$[\hat{\theta}, \hat{L}^2] = 2i\hbar\hat{L} \quad (\text{F.15})$$

$$[\hat{L}, \cos\theta] = i\hbar \sin\theta, \quad (\text{F.16})$$

and the results follow. The electric dipole is different because, unlike the magnetic moment, it is not associated with angular momentum. (An electric dipole is just two, opposite charges, which are spatially separated. A magnetic dipole is two, opposite charges, which are orbiting each other and so is associated with angular momentum.) Note that the Hamiltonian contains no dissipation, so the electric dipole oscillates forever, and the magnetic dipole precesses forever. Both dipoles will only line up with the external field if we allow them to change their energy, by exchanging energy with a heat bath. This is a dissipative process.

(2.1)  $-1.65 \times 10^{-15}; 7.8 \times 10^{-12}$

(2.2) A duck is mainly water and has a mass of 2–3 kg. An iron filing might be  $0.1 \text{ mm}^3 = 10^{-10} \text{ m}^3$  (as a guess) and each iron atom carries  $2.2 \mu_B$ . The density of iron is  $7873 \text{ kg m}^{-3}$  and the relative atomic mass is 55.847 g. This can be used to show that the required field is 1 mT (the Earth's field is about 0.05 mT). A cow might be 400 kg, and assuming that it too is made of water means that with about 200 times more mass, and hence more volume, a much smaller field ( $\sim 2 \mu\text{T}$ ) is needed to produce the same level of magnetization. Of course we have ignored the effect of haemoglobin.

(2.3)  $2.74 \times 10^{-5} \text{ JT}^{-1}$

(2.4) Hint: use  $Z = 2 \cosh(\mu_B B / k_B T)$  and  $F = -nk_B T \log Z$ . Then  $E = -n(d \log Z / d\beta)$  and  $C = \partial E / \partial T$ . The entropy can be obtained either using  $S = -(\partial F / \partial T)_B$  or  $S = (E - F) / T$ .

(2.5) Hint: a useful intermediate result to prove is that

$$\frac{\partial \log Z_y}{\partial y} = \mathbf{B}_J(y). \quad (\text{F.17})$$

(2.7) (a)  $^5\text{I}_8$  (b)  $^4\text{I}_{15/2}$  (c)  $^3\text{H}_6$  (d)  $^1\text{S}_0$

(2.8) Hints: (1) You will need to evaluate a term containing  $\mathbf{p} \cdot \mathbf{A} + \mathbf{A} \cdot \mathbf{p}$ . This is proportional to

$$\mathbf{p} \cdot \boldsymbol{\mu} \times \mathbf{r} + \boldsymbol{\mu} \times \mathbf{r} \cdot \mathbf{p} = 2\mathbf{r} \times \mathbf{p} \cdot \boldsymbol{\mu}, \quad (\text{F.18})$$

where the final equality works because  $\boldsymbol{\mu}$  does not depend on position, so commutes with  $\mathbf{r}$  and  $\mathbf{p}$ , and you can commute  $\mathbf{r}$  and  $\mathbf{p}$  because they appear in a scalar triple product so you are never multiplying the same component of  $\mathbf{r}$  and  $\mathbf{p}$ .

(2)  $\nabla^2(\frac{1}{r}) = \nabla \cdot \nabla(\frac{1}{r}) = 4\pi\delta(\mathbf{r})$ . An analogous result is  $(\mathbf{S} \cdot \nabla)\nabla(\frac{1}{r}) = \mathbf{S} \frac{4\pi}{3}\delta(\mathbf{r})$ , where the factor of 3 is due to the fact that  $\mathbf{S}$  picks out one of out of the three components of the singularity.

(2.9)  $1.89 \times 10^{-4} \text{ emu mol}^{-1}; 9.68 \times 10^{-7} \text{ emu g}^{-1}$

(2.10) At fixed  $T$ ,

$$T \delta S = -\mathbf{B} \cdot \delta \mathbf{M} - \mathbf{M} \cdot \delta \mathbf{B} + \frac{\partial(nk_B T \log Z)}{\partial \mathbf{B}} \cdot \delta \mathbf{B}, \quad (\text{F.19})$$

but using  $\mathbf{M} = nk_B T (\partial \log Z / \partial \mathbf{B})$ , the result follows.

(3.1)  $\text{Sc}^{2+}$  has one 3d electron ( $l = 2, s = \frac{1}{2}$ ) and has 5 orbital states characterized by  $l_z = -2, -1, 0, 1, 2$ . The energy levels are thus given by 0,  $A$ ,  $4A$  (degeneracies 2, 4 and 4 respectively, including spin) with the ground state being twofold degenerate ( $E = 0$ ) if  $A > 0$ , or fourfold degenerate ( $E = 4A$ ) if  $A < 0$ .

Let us write states using the notation  $|l_z, s_z\rangle$ . If  $A > 0$ , the ground state levels are  $|0, \frac{1}{2}\rangle$  and  $|0, -\frac{1}{2}\rangle$  and are not split by the spin-orbit interaction (because  $\lambda l_z s_z = 0$ ). The levels are split by a magnetic field in any direction, and lead to a Curie-like susceptibility providing  $\mu_B B \ll k_B T \ll A$ . This condition ensures that you do not populate excited states ( $k_B T \ll A$ ) and you keep on the linear part of the Brillouin function ( $\mu_B B \ll k_B T$ ).

If  $A < 0$ , the ground state levels are  $|2, \frac{1}{2}\rangle, |2, -\frac{1}{2}\rangle, |-2, \frac{1}{2}\rangle$  and  $|-2, -\frac{1}{2}\rangle$  and the spin-orbit interaction acts so that  $|2, -\frac{1}{2}\rangle$  and  $|-2, \frac{1}{2}\rangle$  are lowered in energy by  $\lambda$  while  $|2, \frac{1}{2}\rangle$  and  $|-2, -\frac{1}{2}\rangle$  are raised in energy by  $\lambda$ . The lower levels will only now be split by a magnetic field parallel to  $z$  and so the susceptibility will be Curie-like if  $\mu_B B \ll k_B T \ll \lambda$ . The susceptibility will be temperature independent if the field is applied perpendicular to  $z$ .

(3.2) Use

$$\begin{aligned} (1+x)^{-1/2} &= 1 - \frac{x}{2} + \frac{(-\frac{1}{2})(-\frac{3}{2})}{2!}x^2 + \frac{(-\frac{1}{2})(-\frac{3}{2})(-\frac{5}{2})}{3!}x^3 \\ &\quad + \frac{(-\frac{1}{2})(-\frac{3}{2})(-\frac{5}{2})(-\frac{7}{2})}{4!}x^4 + \dots \\ &= 1 - \frac{x}{2} + \frac{3x^2}{8} - \frac{5x^3}{16} + \frac{35x^4}{128} + \dots \end{aligned} \quad (\text{F.20})$$

(3.3) Converting  $V$  into spherical polars produces

$$V = \frac{Dr^4}{8} \left[ \sin^4 \theta (e^{4i\phi} + e^{-4i\phi} + 6) + 8 \cos^2 \theta - \frac{24}{5} \right]. \quad (\text{F.21})$$

For the matrix elements, all off-diagonal terms vanish except  $V_{2,-2}$  and  $V_{-2,2}$ . The evaluation of the non-zero matrix elements are straightforward but tedious. Here is one example:

$$\begin{aligned} V_{1,1} &= \int_0^\infty r^6 R^2(r) dr \frac{D}{8} \int_0^\pi \left( 6 \sin^2 \theta + 8 \cos^4 \theta - \frac{24}{5} \right) \\ &\quad \times 4 \sin^3 \theta \cos^2 \theta d\theta \int_0^{2\pi} d\phi \\ &= -4A. \end{aligned} \quad (\text{F.22})$$

The eigenvalues and eigenvectors are

| Eigenvalue                                                   | Eigenvector                                            |
|--------------------------------------------------------------|--------------------------------------------------------|
| $A + 5A \left( 1 + \frac{4\mu_B^2 B^2}{25A^2} \right)^{1/2}$ | $\frac{( 2\rangle + x_+  -2\rangle)}{(1+x_+^2)^{1/2}}$ |
| $-4A + \mu_B B$                                              | $ 1\rangle$                                            |
| $6A$                                                         | $ 0\rangle$                                            |
| $-4A - \mu_B B$                                              | $ -1\rangle$                                           |
| $A - 5A \left( 1 + \frac{4\mu_B^2 B^2}{25A^2} \right)^{1/2}$ | $\frac{( 2\rangle + x_-  -2\rangle)}{(1+x_-^2)^{1/2}}$ |

where

$$x_{\pm} = \pm \left( 1 + \left( \frac{2\mu_B B}{5A} \right)^2 \right)^{1/2} - \frac{2\mu_B B}{5A}. \quad (\text{F.23})$$

Hence in the limit of low field, the eigenvalues and eigenvectors are

| Eigenvalue                      | Eigenvector                                 |
|---------------------------------|---------------------------------------------|
| $6A + \frac{2\mu_B^2 B^2}{5A}$  | $\frac{( 2\rangle +  -2\rangle)}{\sqrt{2}}$ |
| $-4A + \mu_B B$                 | $ 1\rangle$                                 |
| $6A$                            | $ 0\rangle$                                 |
| $-4A - \mu_B B$                 | $ -1\rangle$                                |
| $-4A - \frac{2\mu_B^2 B^2}{5A}$ | $\frac{( 2\rangle -  -2\rangle)}{\sqrt{2}}$ |

and so if  $k_B T \ll A$ , then you only populate the lowest levels which are a triplet if  $A > 0$  or a doublet if  $A < 0$ . Considering these lowest levels only, the partition function can be written

$$Z = \begin{cases} 1 + e^{-\mu_B B / k_B T} + e^{\mu_B B / k_B T} & A > 0 \\ 1 + e^{-2\mu_B^2 B^2 / 5A k_B T} & A < 0 \end{cases}, \quad (\text{F.24})$$

and the results follow using  $F = -nk_B T \log Z$  and  $\mathbf{M} = -\partial F / \partial \mathbf{B}$ .

(3.4)  $I = \frac{3}{2}$ ,  $A = 2 \times 10^{-6}$  eV. Not much can be said about the nuclear moment. Line broadening due to phonons is large at high temperature. Low temperatures are often needed to resolve fine structure.

(3.5) Measure  $g_J J = 7$  (for  $\text{Gd}^{3+}$ ), 5 (for  $\text{Fe}^{3+}$ ) and 3 (for  $\text{Cr}^{3+}$ ). This agrees with the predictions of Hund's rules for all of them except  $\text{Cr}^{3+}$  for which one would expect  $J = \frac{3}{2}$  and  $g_J = \frac{2}{5}$ . This is because orbital quenching occurs and means that  $g_J = 2$ . (Orbital quenching also occurs for  $\text{Fe}^{3+}$  but you do not notice because  $L = 0$  is predicted from Hund's rules anyway.)

(3.6) The fields for 60 MHz are

|                 |        |
|-----------------|--------|
| $^1\text{H}$    | 1.41 T |
| $^2\text{H}$    | 9.18 T |
| $^{13}\text{C}$ | 5.60 T |
| $^{19}\text{F}$ | 1.50 T |

(3.7) 0.322 T

(3.8) First derive matrix representations for  $\hat{S}_z$  and  $\hat{S}_x$  for an  $S = 1$  particle:

$$\hat{S}_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \quad (\text{F.25})$$

$$\hat{S}_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}. \quad (\text{F.26})$$

Hence, the Hamiltonian is

$$\hat{\mathcal{H}} = \begin{pmatrix} D + g_{\parallel}\mu_B B \cos\theta & \frac{g_{\perp}\mu_B B \sin\theta}{\sqrt{2}} & 0 \\ \frac{g_{\perp}\mu_B B \sin\theta}{\sqrt{2}} & 0 & \frac{g_{\perp}\mu_B B \sin\theta}{\sqrt{2}} \\ 0 & \frac{g_{\perp}\mu_B B \sin\theta}{\sqrt{2}} & D - g_{\parallel}\mu_B B \cos\theta \end{pmatrix} \quad (\text{F.27})$$

and the eigenvalues can be found to give the required result. When  $\theta = 0$  the eigenvalues simplify to

$$E = 0, D \pm g_{\parallel}\mu_B B. \quad (\text{F.28})$$

When  $\theta = \pi/2$  the eigenvalues simplify to

$$E = D, \frac{D}{2} \pm \sqrt{\left(\frac{D}{2}\right)^2 + (g_{\perp}\mu_B B)^2}. \quad (\text{F.29})$$

(3.9)  $82.4 \text{ ms}^{-1}$ ;  $7.7 \times 10^{-19} \text{ ms}^{-1}$ ;  $9.4 \times 10^{14} \text{ Hz}$ ;  $9.1 \times 10^{-9} \text{ Hz}$ ;  $2.3 \times 10^7 \text{ Hz}$ ;  $0.1 \text{ } \mu\text{eV}$ ;  $7.7 \times 10^{-4} \text{ cm}^{-1}$ .

(3.10) If the lowest frequency you can easily measure has a period of  $20 \text{ } \mu\text{s}$ , it has a frequency of 50 kHz, corresponding to a field of about  $4 \times 10^{-4} \text{ T}$ . The fraction of muons living for  $20 \text{ } \mu\text{s}$  or longer is  $e^{-20/2.2} = 1.1 \times 10^{-4}$  corresponding to just over  $10^3$  of the  $10^7$  muons implanted.

This result follows from the exponential decay of muons with mean lifetime  $\tau_\mu$ . The fraction living longer than  $T$  is given by

$$\frac{\int_T^\infty e^{-t/\tau_\mu} dt}{\int_0^\infty e^{-t/\tau_\mu} dt} = e^{-T/\tau_\mu}. \quad (\text{F.30})$$

If the pulse-width is 50 ns, destructive interference can occur if the spin-precession period is 100 ns, corresponding to a frequency of 10 MHz, giving an upper limit on the field of  $\sim 0.07 \text{ T}$ .

If  $B = 0.4 \text{ T}$ ,  $f = 54.2 \text{ MHz}$ .

(3.11) For  $\text{Fe}^{3+}$ ,  $J = \frac{5}{2}$  and  $g_J = 2$ . The saturation moment is  $g_J J \mu_B = 5\mu_B$ . Susceptibility measures  $\mu_{\text{eff}}^2 = g_J^2 J(J+1)\mu_B^2$  and hence

$$g_J \sqrt{J(J+1)} \mu_B = 2 \times \sqrt{\frac{5}{2} \times \frac{7}{2}} \mu_B = \sqrt{35} \mu_B = 5.92 \mu_B. \quad (\text{F.31})$$

The saturation moment involves  $\hat{J}_z$  because it is a measurement of the moment saturated along a particular direction. In contrast, susceptibility involves  $\hat{J}^2$ .

(4.1) There are several ways to derive this result. One way is to start with the vector potential from dipole  $\mathbf{l}$  which is

$$\mathbf{A}_1 = \frac{\mu_0}{4\pi} \nabla \times \frac{\boldsymbol{\mu}_1}{r}, \quad (\text{F.32})$$

and this produces a magnetic field

$$\mathbf{B}_1 = \nabla \times \mathbf{A}_1 = \frac{\mu_0}{4\pi} \left[ \nabla \left( \nabla \cdot \frac{\boldsymbol{\mu}_1}{r} \right) - \nabla^2 \left( \frac{\boldsymbol{\mu}_1}{r} \right) \right], \quad (\text{F.33})$$

and gives rise to an energy

$$\begin{aligned} E &= -\boldsymbol{\mu}_2 \cdot \mathbf{B}_1 \\ &= -\frac{\mu_0}{4\pi} \left[ (\boldsymbol{\mu}_1 \cdot \nabla)(\boldsymbol{\mu}_2 \cdot \nabla) \frac{1}{r} - (\boldsymbol{\mu}_1 \cdot \boldsymbol{\mu}_2) \nabla^2 \frac{1}{r} \right], \\ &= \frac{\mu_0}{4\pi r^3} \left[ \boldsymbol{\mu}_1 \cdot \boldsymbol{\mu}_2 - \frac{3(\boldsymbol{\mu}_1 \cdot \mathbf{r})(\boldsymbol{\mu}_2 \cdot \mathbf{r})}{r^2} \right]. \end{aligned} \quad (\text{F.34})$$

It is also possible to derive the result by using the magnetic scalar potential  $\phi_1 = \boldsymbol{\mu}_1 \cdot \mathbf{r}/4\pi r^3$  and then  $\mathbf{B}_1 = -\mu_0 \nabla \phi_1$ .

(4.2) 1 Å: (a) 1.855 T (b) 1.855 mT; 10 Å: (a) 0.927 T (b) 0.927 mT.

(4.3) The lattice constant in Fe (which is bcc) is  $a = 2.87 \text{ Å}$ . The nearest neighbour distance is therefore  $a/\sqrt{2}$  and so the dipole-dipole energy is  $\sim (\mu_0 \times (2.2\mu_B)^2/4\pi r^3) = 30 \text{ } \mu\text{eV}$ .  $J \approx k_B T_C \sim 0.09 \text{ eV}$ , which is about 3000 times larger.

(4.4) The bandwidth  $\sim 0.05 \text{ eV}$  and the Coulomb energy  $\sim 1 \text{ eV}$ .

$$J \sim -\frac{t^2}{U} = -2.5 \text{ meV}.$$

Hence if  $|J| = k_B T_N$ , then  $T_N = 29 \text{ K}$ .

(4.5) The partition function is

$$Z = 1 + e^{-\beta \Delta} (e^{\beta g \mu_B B} + 1 + e^{-\beta g \mu_B B}). \quad (\text{F.35})$$

This can be used to derive  $F$  and hence  $M$  given by

$$M = \frac{2n\beta(g\mu_B)^2 B}{3 + e^{\beta \Delta}}, \quad (\text{F.36})$$

and the result for  $\chi \ll 1$  then follows. Note that at high temperatures  $T \gg \Delta/k_B$ ,  $\chi \rightarrow n\mu_0 g^2 \mu_B^2/2k_B T$  which is a Curie law. It corresponds to all levels occupied, so that the susceptibility is  $\frac{3}{4}$  of an  $S = 1$  system (for which  $S(S+1) = 2$ ) and  $\frac{1}{4}$  of an  $S = 1$  system (for which  $S(S+1) = 0$ ) so is  $\frac{3}{2}$  of a classical Curie law.

At low temperatures,  $T \ll \Delta/k_B$ , then if  $\Delta > 0$  one has  $\chi \propto e^{-\Delta/k_B T} \rightarrow 0$  (non-magnetic because only the singlet is occupied), but if  $\Delta < 0$  one has  $\chi \rightarrow 2n\mu_0 g^2 \mu_B^2/3k_B T$  (corresponding to the occupied triplet,  $S(S+1) = 2$ , and so twice a classical Curie law).

(4.6) Hint: First show that

$$\boldsymbol{\mu}_{11} \cdot \boldsymbol{\mu}_{12} \propto \hat{I}_{1z} \hat{I}_{2z} + \frac{1}{2} (\hat{I}_{1+} \hat{I}_{2-} + \hat{I}_{1-} \hat{I}_{2+}) \quad (\text{F.37})$$

and using  $\mathbf{r} = r(\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta)$ , show that

$$\boldsymbol{\mu}_I \cdot \hat{\mathbf{r}} \propto \cos\theta \hat{I}_z + \frac{1}{2} \sin\theta [e^{-i\phi} \hat{I}_+ + e^{i\phi} \hat{I}_-]. \quad (\text{F.38})$$

The results then follow fairly quickly.

- (4.8) Note that if  $\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2 + \mathbf{S}_3$  then

$$\mathbf{S}^2 = \mathbf{S}_1^2 + \mathbf{S}_2^2 + \mathbf{S}_3^2 + 2(\mathbf{S}_1 \cdot \mathbf{S}_2 + \mathbf{S}_2 \cdot \mathbf{S}_3 + \mathbf{S}_3 \cdot \mathbf{S}_1). \quad (\text{F.39})$$

Hence the Hamiltonian can be rewritten in terms of  $\mathbf{S}^2$ ,  $\mathbf{S}_1^2$ ,  $\mathbf{S}_2^2$  and  $\mathbf{S}_3^2$  giving  $E = J(6 - S(S+1))$ . Hence for the possible values of  $S$  the required energies are obtained.

- (5.1)  $B_{\text{mf}} = 2.1 \times 10^3$  T. The number of atoms per unit volume in Fe are  $n = 8.49 \times 10^{28} \text{ m}^{-3}$ , and using  $M = n \cdot 2.2\mu_B$  then  $\mu_0 M = 2.2$  T. Hence  $B_{\text{mf}}$  is  $10^3$  times bigger than  $\mu_0 M$ .

- (5.2) The magnetization is given by

$$\frac{M}{M_s} = B_S(y) = \alpha y - \beta y^3 + \dots, \quad (\text{F.40})$$

where  $\alpha$  and  $\beta$  are constants and

$$y = \frac{\gamma M}{TM_s}, \quad (\text{F.41})$$

where  $\gamma$  is a positive constant given by  $T_C/\alpha$ . Near  $T_C$ , these equations lead to

$$\left(\frac{M}{M_s}\right)^2 \approx \frac{\alpha^3}{\beta} \frac{(T_C - T)}{T_C} \quad (\text{F.42})$$

and this leads to the required result using the values of  $\alpha$  and  $\beta$  that can be obtained from eqn 5.38.

The energy is given by  $E = -\frac{1}{2}\lambda M^2$  and using eqn F.42 one can show that at  $T_C$

$$\frac{d(M^2)}{dT} = -\frac{M_s^2 \alpha^3}{\beta T_C}, \quad (\text{F.43})$$

and hence the result for the specific heat can be found.

In general one can now easily show that

$$C = \begin{cases} \frac{5}{2}nk_B \left[ \frac{(2S+1)^2-1}{(2S+1)^2+1} \right] \left( 3\left(\frac{T}{T_C}\right)^2 - 2\left(\frac{T}{T_C}\right) \right) & T \leq T_C \\ 0 & T > T_C. \end{cases} \quad (\text{F.44})$$

- (5.4) The energy is given by

$$E = -2NS^2(J_1 \cos \theta + J_2 \cos 2\theta), \quad (\text{F.45})$$

and energy minimization  $\partial E/\partial \theta = 0$  implies that either  $\sin \theta = 0$  (and hence  $\theta = 0$  or  $\pi$ ) or  $\cos \theta = -J_1/4J_2$ . These solutions are ferromagnetic ( $\theta = 0$ ), antiferromagnetic ( $\theta = \pi$ ) and helimagnetic ( $\cos \theta = -J_1/4J_2$ ) and substitution back into the equation for the energy reproduces eqn 5.45. The helimagnetic solution is only possible if  $|J_1| < 4|J_2|$ .

Which solution gives the lowest energy can be evaluated by comparing the energies of each solution. Alternatively one can look at the stability of each solution by evaluating  $\partial^2 E/\partial \theta^2$ .

- (5.5) When  $y \gg 1$ ,

$$\tanh y = \frac{e^y - e^{-y}}{e^y + e^{-y}} = \frac{1 - e^{-2y}}{1 + e^{-2y}} \approx 1 - 2e^{-2y}. \quad (\text{F.46})$$

This can be used together with

$$\frac{M}{M_s} = \tanh y, \quad (\text{F.47})$$

and also with the fact that near  $T_C$  one has

$$\frac{M}{M_s} = \frac{T}{T_C} y, \quad (\text{F.48})$$

to derive the final result.

- (5.6) Recall the mean-field theory for a ferromagnet. In that case the magnetization is given by

$$M = \frac{C}{\mu_0 T} (\lambda M + \mu_0 H), \quad (\text{F.49})$$

where  $C$  is the Curie constant. This rearranges to

$$M \left( 1 - \frac{\lambda C}{\mu_0 T} \right) = \frac{HC}{T}, \quad (\text{F.50})$$

and so the susceptibility is

$$\chi = \frac{M}{H} = \frac{C}{T - \theta} \quad (\text{F.51})$$

where the Weiss temperature is  $\theta = \lambda C/\mu_0$ .

For this ferrimagnet, the magnetization in each sublattice is

$$M_1 = \frac{C_1}{\mu_0 T} (\mu_0 H - \lambda M_2) \quad (\text{F.52})$$

$$M_2 = \frac{C_2}{\mu_0 T} (\mu_0 H - \lambda M_1) \quad (\text{F.53})$$

which becomes

$$\begin{pmatrix} T & \lambda C_1/\mu_0 \\ \lambda C_2/\mu_0 & T \end{pmatrix} \begin{pmatrix} M_1 \\ M_2 \end{pmatrix} = H \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}. \quad (\text{F.54})$$

The determinant of the matrix in this equation is therefore zero if  $T^2 - \lambda^2 C_1 C_2/\mu_0 = 0$  and hence one can deduce that the paramagnetic regime is when  $T > \theta$  where  $\theta = \lambda(C_1 C_2)^{1/2}/\mu_0$ . The susceptibility is given by  $\chi = (M_1 + M_2)/H$  which can be found by inverting eqn F.54 to give

$$\begin{pmatrix} M_1 \\ M_2 \end{pmatrix} = \frac{H}{T^2 - \theta^2} \begin{pmatrix} T & -\lambda C_1/\mu_0 \\ -\lambda C_2/\mu_0 & T \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} \quad (\text{F.55})$$

and hence the final answer.

- (5.7) The first part follows straightforwardly from an application of  $\hat{S}^\pm = \hat{S}^x \pm i\hat{S}^y$ .

For the Heisenberg ferromagnet, the term  $\hat{S}_i^z \hat{S}_j^z$  operating on the ground state produces  $S^2$  and the ground state is an eigenstate of  $\hat{S}_i^z \hat{S}_j^z$ . There are  $N$  such terms and hence  $E = -NS^2$ . The terms  $\hat{S}_i^+ \hat{S}_j^-$  and  $\hat{S}_i^- \hat{S}_j^+$  produce nothing because in each case the raising operator annihilates the ground state. Thus the ground state is an eigenstate of both  $\hat{S}_i^+ \hat{S}_j^-$  and  $\hat{S}_i^- \hat{S}_j^+$  with eigenvalue zero.

For the Heisenberg antiferromagnet the two terms  $\hat{S}_i^+ \hat{S}_j^-$  and  $\hat{S}_i^- \hat{S}_j^+$  produce states other than the 'obvious' ground state. Therefore this 'obvious' ground state is not an eigenstate of the system.

- (5.8) The range of possible momentum transfer is  $0 \text{ nm}^{-1}$  to  $\sqrt{k_{\text{in}}} = 2\sqrt{2}\pi/\lambda = 29.6 \text{ nm}^{-1}$ .

(a) At 100 K, just get the chemical Bragg reflections at

$$Q = \left[ \left( \frac{2\pi}{a} \right)^2 (h^2 + k^2) + \left( \frac{2\pi}{c} \right)^2 l^2 \right]^{1/2}. \quad (\text{F.56})$$

The symmetry is bcc, so reflections are seen when  $h + k + l$  is even.

(b) At 10 K, also see the magnetic Bragg reflections.

The observed reflections for both cases are listed below. The magnitude of the scattering vector  $Q$  are listed, together with  $2\theta$ , the angle through which the incident beam is scattered. The magnetic reflections are indicated by an asterisk and the multiplicity,  $M$ , of each reflection is given in the final column.

| $h$ | $k$ | $l$ | $Q$<br>(nm <sup>-1</sup> ) | $2\theta$<br>(°) |   | $M$ |
|-----|-----|-----|----------------------------|------------------|---|-----|
| 0   | 1   | 0   | 12.566                     | 34.915           | * | 4   |
| 1   | 0   | 0   | 12.566                     | 34.915           | * |     |
| 1   | 1   | 0   | 17.772                     | 50.208           |   | 4   |
| 0   | 0   | 1   | 20.944                     | 60.000           | * | 2   |
| 0   | 1   | 1   | 24.425                     | 71.337           |   | 8   |
| 1   | 0   | 1   | 24.425                     | 71.337           |   |     |
| 0   | 2   | 0   | 25.133                     | 73.740           |   | 4   |
| 2   | 0   | 0   | 25.133                     | 73.740           |   |     |
| 1   | 1   | 1   | 27.468                     | 81.952           | * | 8   |
| 1   | 2   | 0   | 28.099                     | 84.261           | * | 8   |
| 2   | 1   | 0   | 28.099                     | 84.261           | * |     |

- (5.9) (a) The first expression for  $\tilde{m}$  follows from the definition of  $\tanh y$ . This can be rearranged to give

$$x^2 = \frac{1 + \tilde{m}}{1 - \tilde{m}}, \quad (\text{F.57})$$

and taking the log of this equation gives eqn 5.57. Using eqn F.48 leads to eqn 5.58.

- (b) Rearranging eqn 5.60 to make  $x$  the subject leads to

$$(1 - \tilde{m})x^2 - \tilde{m}x - (1 + \tilde{m}) = 0, \quad (\text{F.58})$$

which is a quadratic. This has real solutions if  $4 - 3\tilde{m}^2 \geq 0$  and are simply found to be eqn 5.62. Now  $1 \leq 4 - 3\tilde{m}^2 \leq 2$  if  $0 \leq \tilde{m} \leq 1$ , so for  $x > 0$  we need the positive solution in eqn 5.62. This then yields

$$y = \log(\tilde{m} + \sqrt{4 - 3\tilde{m}^2}) - \log(2(1 - \tilde{m})), \quad (\text{F.59})$$

and in this case one finds  $y = 3\tilde{m}T_C/2T$  and the final result follows.

- (6.1) The average energy is found to be

$$\langle E \rangle = -\frac{NJ}{4} \tanh\left(\frac{J}{2k_B T}\right), \quad (\text{F.60})$$

and the heat capacity per spin is then

$$C = \frac{J^2}{4k_B T^2 \cosh^2(J/2k_B T)}. \quad (\text{F.61})$$

This behaves like  $J^2/4k_B T^2$  at high temperature, and as  $J^2 e^{-J/k_B T}/4k_B T^2$  at low temperature. It is thus zero at both  $T = 0$  and  $T = \infty$  and exhibits a broad maximum in between. There is thus no cusp and this reinforces the point that there are no phase transitions in purely one-dimensional models.

- (6.2) (b)  $\hbar\omega(\mathbf{q}) = S[J(0) - J(\mathbf{q}) + K(0)]$ .

(c) (i)  $\hbar\omega(\mathbf{q}) = 2SJ_0(1 - \cos qa) + 2SK_0$ .

(ii)  $\hbar\omega(\mathbf{q}) = 2SJ_0(2 - \cos q_x a - \cos q_y a) + 4SK_0$ .

(iii)

$$\hbar\omega(\mathbf{q}) = 8SJ_0 \left(1 - \cos \frac{q_x a}{2} \cos \frac{q_y a}{2} \cos \frac{q_z a}{2}\right) + 8SK_0.$$

- (6.3) At small  $q$  one can use  $\cos qa \approx 1 - (qa)^2/2$  and obtain

(i)  $\hbar\omega = S(J_0 a^2 q^2 + 2K_0)$

(ii)  $\hbar\omega = S(J_0 a^2 q^2 + 4K_0)$

(iii)  $\hbar\omega = S(J_0 a^2 q^2 + 8K_0)$

In the Ising case ( $J_0 = 0$ ), one has  $\hbar\omega = 2dSK_0 \equiv \Delta$  where  $d = 1, 2, 4$  for cases (i), (ii) and (iii) respectively. In this model, spin waves 'cost' the same energy  $\Delta$  regardless of wave vector. The number of spin waves is then proportional to  $e^{-\Delta/k_B T}$ , and so their energy is proportional to  $\Delta e^{-\Delta/k_B T}$ . The specific heat is then easily found to be proportional to

$$\frac{\Delta^2}{k_B T^2} e^{-\Delta/k_B T},$$

in agreement with Exercise 6.1 with  $\Delta$  in this problem being equal to  $J$  in that problem.

In the Heisenberg case ( $K_0 = 0$ ),  $\hbar\omega = Dq^2$  where the spin-stiffness  $D = SJ_0 a^2$ . The number of spin waves  $N_s$  is given by

$$N_s \propto \int \frac{d^n q}{e^{Dq^2/k_B T} - 1}, \quad (\text{F.62})$$

and at small  $q$

$$\frac{1}{e^{Dq^2/k_B T} - 1} \approx \frac{k_B T}{Dq^2}. \quad (\text{F.63})$$

The integral therefore diverges in one or two dimensions (where  $d^n q$  is equal to  $dq$  or  $2\pi q dq$  respectively), but converges for three dimensions (where  $d^n q$  is equal to  $4\pi q^2 dq$ ). No long range order is therefore possible for the pure (isotropic) Heisenberg model in one or two dimensions.

- (6.4) In the antiferromagnetic case, one can follow the derivation for the ferromagnetic case, but you have to remember which sublattice you are on. One finds that

$$\begin{aligned} \hbar\dot{S}_j^+ &= 2iJS[2S_j^+ + S_{j-1}^+ + S_{j+1}^+] & j \text{ odd} \\ \hbar\dot{S}_j^+ &= -2iJS[2S_j^+ + S_{j-1}^+ + S_{j+1}^+] & j \text{ even.} \end{aligned} \quad (\text{F.64})$$

Hence substituting in

$$S_j^+ = \begin{cases} ue^{i(qja - \omega t)} & j \text{ odd} \\ ve^{i(qja - \omega t)} & j \text{ even,} \end{cases} \quad (\text{F.65})$$

one finds that

$$i\hbar\omega \begin{pmatrix} u \\ v \end{pmatrix} = 4iJS \begin{pmatrix} 1 & \cos qa \\ -\cos qa & -1 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}, \quad (\text{F.66})$$

and hence

$$\begin{vmatrix} 1 - \frac{\hbar\omega}{4JS} & \cos qa \\ \cos qa & 1 + \frac{\hbar\omega}{4JS} \end{vmatrix} = 0 \quad (\text{F.67})$$

so that  $\hbar\omega = 4JS \sin qa$ .

- (6.5) Using  $\partial F/\partial M = 0$ , some rearrangement quickly yields

$$M^2 = -\frac{2a_0(T - T_C)}{4b} + \frac{\mu_0 H}{4bM}, \quad (\text{F.68})$$

which is of the required form. Plotting  $M^2$  against  $M/H$  gives a straight-line, with the intercept on the  $M^2$  axis changing sign at  $T = T_C$ .

(6.6) The model in appropriate units is

$$\epsilon = \frac{1}{2} \sin^2(\theta - \phi) - h \cos \phi. \quad (\text{F.69})$$

For  $\theta = 0$ , energy minimization gives

$$\sin \phi (\cos \phi + h) = 0 \quad (\text{F.70})$$

for which the solutions are  $\phi = 0, \pi, \cos^{-1}(-h)$ . Evaluation of  $\partial^2 E / \partial \theta^2$  can be used to examine the stability of these solutions.

For  $\theta = \pi/2$ , energy minimization gives

$$\sin \phi (h - \cos \phi) = 0 \quad (\text{F.71})$$

for which the solutions are  $\phi = 0, \pi, \cos^{-1}(h)$ .

(6.7) The components of magnetization are

$$M_x = |\mathbf{M}| \sin \frac{\pi r}{R} \quad (\text{F.72})$$

$$M_y = |\mathbf{M}| \cos \frac{\pi r}{R}, \quad (\text{F.73})$$

and using results such as

$$\frac{dr}{dx} = \frac{d}{dx} \sqrt{x^2 + y^2 + z^2} = \frac{x}{r} \quad (\text{F.74})$$

one can evaluate all the partial differential of  $M_x$  and  $M_y$ . Hence

$$(\nabla M_x)^2 + (\nabla M_y)^2 = |\mathbf{M}|^2 \frac{\pi^2}{R^2}, \quad (\text{F.75})$$

from which the final result follows.

(6.8) 25 Gbits in<sup>-2</sup> is  $3.9 \times 10^{13}$  bits m<sup>-2</sup> in sensible units.

(a) Taking  $\pi a_0^2$  to be the area of a hydrogen atom gives  $3.4 \times 10^{-7}$  bits (hydrogen atom area)<sup>-1</sup>.

(b) Most Shakespearian plays are 20000–40000 words long; with maybe an average 5 letters per word, and about 40 plays, one gets about 4 Mbytes or 32 Mbits (8 bits per byte). Taking a UK stamp, I calculated about 600 copies of the complete works per stamp, although that could be easily doubled with data-compression. “I am ill at all these numbers” (Hamlet Act 2 Scene 2)

DNA stores 1 bit per base pair, which is about 1 bit per cubic nanometre. If you made a sheet of this, this would be equivalent to about  $6 \times 10^5$  Gbits in<sup>-2</sup>, but the great advantage of DNA is that it can coil and is a volume storage technique.

(6.9) The best way to tackle this is to put the magnetic field parallel to  $-z$  which you can do without loss of generality. (Why  $-z$ ? So that the ground state has all its spins aligned parallel to  $z$ . Electrons have negative charge and so their spins are antiparallel to their magnetic moments and will therefore align antiparallel to the applied field.) You should find that each magnon adds an energy  $+g\mu_B B$ .

(6.11) The equation

$$\langle q|i\rangle = \frac{1}{\sqrt{N}} \sum_j e^{-iq \cdot \mathbf{R}_j} \langle j|i\rangle \quad (\text{F.76})$$

can be simplified using  $\langle j|i\rangle = \delta_{ij}$ . The result then follows.

Hint for the next part:  $S^+$  kills the state it is operating on unless the spin it is operating on is  $< S$ .

The exercise shows that each spin has a small transverse component which is perpendicular to the direction of magnetization. The transverse components vary in space in just the manner depicted in Fig. 6.13.

(6.12) It is necessary to solve

$$\Delta E = 0 = \int_{-\infty}^{\infty} \left[ 2A \frac{\partial \theta}{\partial z} \frac{\partial}{\partial z} (\delta \theta) + \frac{\partial f(\theta)}{\partial \theta} \delta \theta \right] dz \quad (\text{F.77})$$

The first part of this integral can be evaluated by parts, yielding

$$\frac{\partial f}{\partial \theta} - 2A \frac{\partial^2 f}{\partial z^2} = 0. \quad (\text{F.78})$$

The question carefully guides the reader through the rest.

(6.13) The wall energy per unit volume due to the length of walls in the structure and is therefore approximately  $\sigma_w L/D$ . The spins inside the small triangles cost an anisotropy energy. The area of one of these triangles is  $D^2/4$  and there are two of them in every horizontal length  $D$ , so their total cost is  $2 \times K \times (D^2/4) \times (1/D) = KD/2$ . Summing these terms and differentiating yields the required result.  $D = 20 \mu\text{m}$ .

(7.1) Using  $g(E_F) = 3n/2E_F$ , the first result follows quite quickly.

To consider the temperature dependence, it is worthwhile to review some mathematical details concerning the Fermi function. Consider  $h(E) = \int_0^E g(E') dE'$ , so that  $g(E) = dh/dE$  and consider the integral

$$\begin{aligned} I &= \int_0^{\infty} \frac{dh}{dE} f(E) dE \\ &= [f(E)g(E)]_0^{\infty} - \int_0^{\infty} h(E) \frac{df}{dE} dE \\ &= - \int_0^{\infty} h(E) \frac{df}{dE} dE. \end{aligned} \quad (\text{F.79})$$

Now put  $x = (E - \mu)/k_B T$  and hence

$$\frac{df}{dE} = -\frac{1}{k_B T} \frac{e^x}{(e^x + 1)^2}. \quad (\text{F.80})$$

Writing

$$h(E) = \sum_{s=0}^{\infty} \frac{x^s}{s!} \left( \frac{d^s h}{dx^s} \right)_{x=0}, \quad (\text{F.81})$$

we have that

$$I = \sum_{s=0}^{\infty} \frac{1}{s!} \left( \frac{d^s h}{dx^s} \right)_{x=0} \int_{-E_F/k_B T}^{\infty} \frac{x^s e^x dx}{(e^x + 1)^2}. \quad (\text{F.82})$$

The integral part of this can be simplified by replacing the lower limit by  $-\infty$ . It vanishes for odd  $s$ , but for even  $s$

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{x^s e^x dx}{(e^x + 1)^2} &= 2 \int_0^{\infty} \frac{x^s e^x dx}{(e^x + 1)^2} \\ &= 2 \int_0^{\infty} dx \sum_{n=0}^{\infty} e^x x^s \\ &\quad \times \left[ (n+1)(-1)^{n+1} e^{-nx} \right] \\ &= 2 \sum_{n=1}^{\infty} (-1)^{n+1} n \int_0^{\infty} x^s e^{-nx} dx \\ &= 2(s!) \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^s} \\ &= 2(s!)(1 - 2^{1-s})\zeta(s), \end{aligned} \quad (\text{F.83})$$

where  $\zeta(s)$  is the Riemann zeta function. ( $\zeta(0) = -\frac{1}{2}$ ,  $\zeta(2) = \frac{\pi^2}{6}$ ,  $\zeta(4) = \frac{\pi^4}{90}$ ).

Thus the integral is

$$\begin{aligned} I &= \sum_{s=0, s \text{ even}}^{\infty} 2 \left( \frac{d^s h}{dx^s} \right)_{x=0} (1 - 2^{1-s}) \zeta(s) \\ &= h + \frac{\pi^2}{6} \left( \frac{d^2 h}{dx^2} \right)_{x=0} + \frac{7\pi^4}{360} \left( \frac{d^4 h}{dx^4} \right)_{x=0} + \dots \\ &= \int_{-\infty}^{\mu} g(E) dE + \frac{\pi^2}{6} (k_B T)^2 \left( \frac{dg}{dE} \right)_{E=\mu} \\ &\quad + \frac{7\pi^4}{360} (k_B T)^4 \left( \frac{d^3 g}{dE^3} \right)_{E=\mu} + \dots \quad (\text{F.84}) \end{aligned}$$

This implies that the number density is

$$n = \int_0^{\mu} g(E) dE + \frac{\pi^2}{6} (k_B T)^2 \left( \frac{dg}{dE} \right)_{E=\mu} + \dots \quad (\text{F.85})$$

Now to first order

$$\int_0^{\mu} g(E) dE = \int_0^{E_F} g(E) dE + (\mu - E_F) g(E_F). \quad (\text{F.86})$$

so for a constant number of electrons,  $n$  takes the same value at all temperatures and so

$$\mu = E_F - \frac{\pi^2}{6} (k_B T)^2 \frac{g'(E_F)}{g(E_F)}. \quad (\text{F.87})$$

The rôle of a magnetic field is then to shift the energy levels by  $\pm \mu_B B$ . The number density for each spin state is given by

$$\begin{aligned} n_{\pm} &= \frac{1}{2} \int_0^{E_F} g(E) dE + (\mu - E_F \mp \mu_B B) g(E_F \mp \mu_B B) \\ &\quad + \frac{\pi^2}{6} (k_B T)^2 \left( \frac{dg}{dE} \right)_{E=\mu \mp \mu_B B} + \dots \quad (\text{F.88}) \end{aligned}$$

and using the approximate relations

$$g(E_F \mp \mu_B B) = g(E_F) \mp \mu_B B \left( \frac{dg}{dE} \right)_{E=\mu} \quad (\text{F.89})$$

and

$$\left( \frac{dg}{dE} \right)_{E=\mu \mp \mu_B B} = \left( \frac{dg}{dE} \right)_{E=\mu} \mp \mu_B B \left( \frac{d^2 g}{dE^2} \right)_{E=\mu} \quad (\text{F.90})$$

one then has

$$n_+ - n_- = -\frac{\pi^2}{6} (k_B T)^2 \mu_B B \left[ \left( \frac{g'}{g} \right)^2 - \frac{g''}{g} \right] + \mu_B B g(E_F). \quad (\text{F.91})$$

Hence the susceptibility is

$$\chi = \mu_0 \mu_B g(E_F) \left[ 1 - \frac{\pi^2}{6} (k_B T)^2 \left[ \left( \frac{g'}{g} \right)^2 - \frac{g''}{g} \right] \right]. \quad (\text{F.92})$$

For  $g(E) \propto E^{1/2}$  it is simple to show that

$$\begin{aligned} \frac{g'}{g} &= \frac{1}{2E_F} \\ \left( \frac{g'}{g} \right)^2 - \frac{g''}{g} &= \frac{1}{2E_F^2} \quad (\text{F.93}) \end{aligned}$$

evaluated at the Fermi energy. The final result then follows.

For a metal with  $E_F \sim 10$  eV (e.g. Sn),  $T_F \sim 10^5$  K, the correction is about  $10^{-5}$ .

(7.2) (a) This is shown in the derivation of eqn 7.23.

(b) The crossover temperature is  $2T_F/3 = 27$  K. The Pauli paramagnetic susceptibility is then  $1.3 \times 10^{-8}$ . The Landau diamagnetic susceptibility is then

$$\chi_L = -\left( \frac{m_e}{m^*} \right)^2 \frac{\chi_P}{3} = -4.3 \times 10^{-7}. \quad (\text{F.94})$$

(7.3) Using  $A_\phi = \frac{1}{2} Br$ , the Schrödinger equation becomes

$$\begin{aligned} &\left[ -\frac{\hbar^2}{2m_e} \left( \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \phi^2} + \frac{\partial^2}{\partial z^2} \right) - \frac{i\hbar e B}{2m_e} \frac{\partial}{\partial \phi} \right. \\ &\quad \left. + \frac{e^2 B^2 r^2}{8m_e} \right] \psi(r, \phi, z) = E \psi(r, \phi, z). \quad (\text{F.95}) \end{aligned}$$

One can then look for solutions of the form

$$\psi(r, \phi, z) = R(r) e^{il\phi} e^{ik_z z} \quad (\text{F.96})$$

and this leads to

$$\begin{aligned} &\left( -\frac{\hbar^2}{2m_e} \left( \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} \right) - \frac{\hbar^2 l^2}{2m_e} + \frac{1}{2} m_e \omega_c^2 r^2 \right. \\ &\quad \left. + \frac{\hbar k_z^2}{2m_e} \right) R(r) = \left( E - \frac{l}{2} \hbar \omega_c \right) R(r). \quad (\text{F.97}) \end{aligned}$$

The solutions are

$$\psi(r, \theta, z) = \exp \left[ i(l\phi + k_z z) - \frac{eBr^2}{4\hbar} \right] r^{|l|} L_{n-1}^{|l|} \left( \frac{eBr^2}{2\hbar} \right), \quad (\text{F.98})$$

where  $L_\alpha^\beta(x)$  is an associated Laguerre polynomial (Appendix C.5), and the energy is

$$E = \left( n + \frac{l}{2} + \left| \frac{l}{2} \right| - \frac{1}{2} \right) \hbar \omega_c + \frac{\hbar k_z^2}{2m_e}. \quad (\text{F.99})$$

In this so-called symmetric gauge, all states circulate about the same point in space, so that rotational symmetry is captured at the expense of translational symmetry.

Hints:

$$\frac{z+1}{z-1} = 1 + \frac{2}{z-1} \quad (\text{F.100})$$

and hence one can show

$$1 + \frac{1-z^2}{2z} \log \left( \frac{z+1}{z-1} \right) = \frac{2}{3z^2} + O\left( \frac{1}{z^3} \right). \quad (\text{F.101})$$

Also,  $\log(-1) = i\pi$ , so that

$$\log \left| \frac{x+1}{x-1} \right| - \log \left( \frac{x+1}{x-1} \right) = i\pi. \quad (\text{F.102})$$

(7.6) (b) It is useful to start with

$$\chi_q = \mu_0 \mu_B^2 \frac{2m_e}{\hbar^2} I$$

where  $I$  is an integral given by

$$I = \int_{|\mathbf{k}| < k_F} g(\mathbf{k}) d^n k \left[ \frac{1}{(k+q)^2 - k^2} + \frac{1}{(k-q)^2 - k^2} \right]. \quad (\text{F.103})$$

In one-dimension

$$\begin{aligned} \int_{|\mathbf{k}| < k_F} \frac{dk}{(k+q)^2 - k^2} &= \int_{-k_F}^{k_F} \frac{dk}{q(q+2k)} \\ &= \frac{1}{2q} \int_{q-2k_F}^{q+2k_F} \frac{du}{u} \\ &= \frac{1}{2q} \int_{|q-2k_F|}^{q+2k_F} \frac{du}{u} \\ &= \frac{1}{2q} \log \left| \frac{q+2k_F}{q-2k_F} \right|. \end{aligned} \quad (\text{F.104})$$

The result then follows.

In two-dimensions, the relevant integral to consider is

$$\begin{aligned} I &= \int_0^{k_F} k dk \int_0^{2\pi} \frac{d\theta}{q(q-2k \cos \theta)} + \frac{d\theta}{q(q+2k \cos \theta)} \\ &= 2 \int_0^{k_F} k dk \int_0^{2\pi} \frac{d\theta}{q^2 - 4k^2 \cos^2 \theta} \end{aligned} \quad (\text{F.105})$$

Now

$$\int_0^{2\pi} \frac{d\theta}{q^2 - 4k^2 \cos^2 \theta} = \begin{cases} \frac{2\pi}{q(q^2 - 4k^2)^{1/2}} & q > 2k_F \\ 0 & q < 2k_F \end{cases} \quad (\text{F.106})$$

so that the integral becomes

$$I = \begin{cases} \pi \left[ 1 - \sqrt{1 - \left( \frac{2k_F}{q} \right)^2} \right] & q > 2k_F \\ 0 & q < 2k_F \end{cases} \quad (\text{F.107})$$

and the final result follows.

# Appendix G

## Symbols, constants and useful equations

|                   |                                                                           |                    |                                                                    |
|-------------------|---------------------------------------------------------------------------|--------------------|--------------------------------------------------------------------|
| $a$               | unit cell dimension                                                       | $\mathbf{H}$       | magnetic field strength (magnetic field)                           |
| $a_0$             | Bohr radius                                                               | $\mathbf{H}_a$     | applied field                                                      |
| $A$               | area                                                                      | $\mathbf{H}_d$     | demagnetizing field                                                |
| $A$               | hyperfine coupling constant                                               | $\mathbf{H}_i$     | internal field                                                     |
| $A$               | continuum exchange constant                                               | $\mathcal{H}$      | Hamiltonian                                                        |
| $A$               | mass number                                                               | $i$                | $= \sqrt{-1}$                                                      |
| $\mathbf{A}$      | magnetic vector potential                                                 | $I$                | current                                                            |
| $b$               | scattering length                                                         | $I$                | nuclear spin quantum number                                        |
| $\mathbf{B}$      | magnetic flux density (magnetic field)                                    | $J$                | total angular momentum quantum number                              |
| $\mathbf{B}_a$    | applied field                                                             | $J$                | exchange constant/integral                                         |
| $\mathbf{B}_d$    | demagnetizing field                                                       | $k$                | wave vector                                                        |
| $\mathbf{B}_i$    | internal field                                                            | $k_B$              | Boltzmann's constant ( $1.3807 \times 10^{-23} \text{ J K}^{-1}$ ) |
| $\mathbf{B}_{mf}$ | molecular field                                                           | $k_F$              | Fermi wave vector                                                  |
| $B_J$             | Brillouin function                                                        | $K$                | Coulomb integral                                                   |
| $c$               | velocity of light in free space ( $2.9979 \times 10^8 \text{ m s}^{-1}$ ) | $K$                | anisotropy constant                                                |
| $D$               | electric displacement                                                     | $K$                | Knight shift                                                       |
| $D$               | axial anisotropy constant                                                 | $K$                | spring constant                                                    |
| $dS$              | surface element                                                           | $K_s$              | surface anisotropy constant                                        |
| $dS$              | entropy change                                                            | $K_v$              | volume anisotropy constant                                         |
| $e$               | $= \exp(1) = 2.718281828$                                                 | $l$                | orbital angular momentum quantum number                            |
| $e$               | magnitude of the electron charge ( $1.6022 \times 10^{-19} \text{ C}$ )   | $l$                | Landau level index                                                 |
| $E$               | energy                                                                    | $m_e$              | electron mass ( $9.109 \times 10^{-31} \text{ kg}$ )               |
| $E_F$             | Fermi energy                                                              | $m_l$              | orbital magnetic quantum number                                    |
| $E_S$             | singlet energy                                                            | $m_I$              | nuclear magnetic quantum number                                    |
| $E_T$             | triplet energy                                                            | $m_J$              | magnetic quantum number                                            |
| $\mathcal{E}$     | electric field                                                            | $m_n$              | neutron mass                                                       |
| $F$               | Helmholtz free energy (Helmholtz function)                                | $m_s$              | spin magnetic quantum number                                       |
| $g(E)$            | density of states in energy space                                         | $\mathbf{M}$       | magnetization (magnetic moment per unit volume)                    |
| $g(k)$            | density of states in wave vector space                                    | $\mathbf{M}_{\pm}$ | magnetization on a sublattice of an antiferromagnet                |
| $g$               | g-factor                                                                  | $\mathbf{M}_s$     | saturation magnetization                                           |
| $g_e$             | electron g-factor                                                         | $n$                | principal quantum number                                           |
| $g_I$             | nuclear g-factor                                                          | $n$                | number of atoms/moments per unit volume                            |
| $g_l$             | g-factor for orbital angular momentum                                     | $\mathbf{N}$       | demagnetizing tensor                                               |
| $g_s$             | g-factor for spin angular momentum                                        | $N$                | demagnetizing factor                                               |
| $g$               | effective g-tensor                                                        | $N$                | number of neutrons in an atomic nucleus                            |
| $\mathbf{G}$      | torque                                                                    | $N_A$              | Avogadro's number                                                  |
| $\mathbf{G}$      | reciprocal lattice vector                                                 | $p$                | dipole moment                                                      |
| $h$               | Planck's constant ( $6.626 \times 10^{-34} \text{ J s}$ )                 | $\mathbf{p}$       | canonical momentum                                                 |
| $\hbar$           | Planck's constant $/2\pi$ ( $1.0546 \times 10^{-34} \text{ J s}$ )        | $\hat{\mathbf{p}}$ | momentum operator                                                  |
|                   |                                                                           | $\mathbf{q}$       | magnon wave vector                                                 |

|                          |                                     |                            |                                                                                |
|--------------------------|-------------------------------------|----------------------------|--------------------------------------------------------------------------------|
| $\mathbf{q}$             | helix wave vector                   | $\epsilon_0$               | electric permittivity of free space ( $8.854 \times 10^{-12} \text{Fm}^{-1}$ ) |
| $\mathbf{Q}$             | scattering vector                   | $\epsilon_r$               | relative dielectric constant                                                   |
| $\mathbf{r}$             | position vector                     | $\theta$                   | angle                                                                          |
| $\hat{\mathbf{r}}$       | position operator                   | $\theta$                   | Weiss temperature                                                              |
| $R$                      | electrical resistance               | $\lambda$                  | wavelength                                                                     |
| $R_e$                    | extraordinary Hall coefficient      | $\lambda$                  | spin-orbit constant                                                            |
| $R_o$                    | ordinary Hall coefficient           | $\lambda_{\text{el-ph}}$   | electron-phonon coupling constant                                              |
| $S$                      | entropy                             | $\lambda_{\text{spin-ph}}$ | spin-phonon coupling constant                                                  |
| $s, S$                   | spin quantum number                 | $\mu$                      | magnetic moment                                                                |
| $\hat{S}$                | spin angular momentum operator      | $\mu_r$                    | relative magnetic permeability                                                 |
| $\hat{S}_+$              | raising operator                    | $\mu_B$                    | Bohr magneton                                                                  |
| $\hat{S}_-$              | lowering operator                   | $\mu_N$                    | nuclear magneton                                                               |
| $\hat{S}$                | effective spin                      | $\mu_{\text{eff}}$         | effective magnetic moment in $\mu_B$ per formula unit                          |
| $t$                      | time                                | $\mu_0$                    | magnetic permeability of free space ( $4\pi \times 10^{-7} \text{H m}^{-1}$ )  |
| $t$                      | hopping integral                    | $\nu$                      | frequency                                                                      |
| $t_p$                    | pulse length                        | $\xi$                      | correlation length                                                             |
| $T$                      | temperature                         | $\rho$                     | resistivity                                                                    |
| $T_c$                    | critical temperature                | $\rho_H$                   | Hall resistivity                                                               |
| $T_C$                    | Curie temperature                   | $\sigma$                   | conductivity                                                                   |
| $T_N$                    | Néel temperature                    | $\sigma$                   | Pauli spin matrices                                                            |
| $T_P$                    | Peierls transition temperature      | $\tau$                     | period                                                                         |
| $T_{SP}$                 | spin-Peierls transition temperature | $\tau$                     | superparamagnetic relaxation time                                              |
| $T_1$                    | spin-lattice relaxation time        | $\phi$                     | angle                                                                          |
| $T_2$                    | spin-spin relaxation time           | $\Phi$                     | magnetic flux                                                                  |
| $T_F$                    | Fermi temperature                   | $\chi$                     | magnetic susceptibility                                                        |
| $U$                      | Coulomb energy                      | $\chi_g$                   | mass susceptibility                                                            |
| $v$                      | velocity                            | $\chi_L$                   | Landau diamagnetic susceptibility                                              |
| $\mathbf{v}$             | velocity vector                     | $\chi_m$                   | molar magnetic susceptibility                                                  |
| $V$                      | potential or potential energy       | $\chi_P$                   | Pauli spin susceptibility                                                      |
| $W$                      | transition rate                     | $\chi_q$                   | $\mathbf{q}$ -dependent susceptibility                                         |
| $Y_{lm_l}(\theta, \phi)$ | spherical harmonic                  | $\chi$                     | spin wave function                                                             |
| $Z$                      | partition function                  | $\psi$                     | wave function                                                                  |
| $Z$                      | atomic number                       | $\Psi_S$                   | singlet wave function                                                          |
| $\alpha$                 | fine structure constant             | $\Psi_T$                   | triplet wave function                                                          |
| $\beta$                  | $= 1/k_B T$                         | $\omega$                   | angular frequency                                                              |
| $\gamma$                 | gyromagnetic ratio                  | $\omega_c$                 | cyclotron frequency                                                            |
| $\gamma_e$               | electron gyromagnetic ratio         | $\omega_L$                 | Larmor precession frequency                                                    |
| $\gamma_\mu$             | muon gyromagnetic ratio             |                            |                                                                                |
| $\gamma_N$               | nuclear gyromagnetic ratio          |                            |                                                                                |
| $\Delta$                 | exchange splitting                  |                            |                                                                                |

**Fundamental constants**

|                                     |                                       |                                                           |
|-------------------------------------|---------------------------------------|-----------------------------------------------------------|
| Bohr radius                         | $a_0$                                 | $5.292 \times 10^{-11} \text{ m}$                         |
| Speed of light in free space        | $c$                                   | $2.9979 \times 10^8 \text{ m s}^{-1}$                     |
| Electronic charge                   | $e$                                   | $1.6022 \times 10^{-19} \text{ C}$                        |
| Planck's constant                   | $h$                                   | $6.626 \times 10^{-34} \text{ J s}$                       |
|                                     | $\hbar/2\pi =$                        | $\hbar$ $1.0546 \times 10^{-34} \text{ J s}$              |
| Boltzmann's constant                | $k_B$                                 | $1.3807 \times 10^{-23} \text{ J K}^{-1}$                 |
| Electron rest mass                  | $m_e$                                 | $9.109 \times 10^{-31} \text{ kg}$                        |
| Proton rest mass                    | $m_p$                                 | $1.6726 \times 10^{-27} \text{ kg}$                       |
| Avogadro's number                   | $N_A$                                 | $6.022 \times 10^{23} \text{ mol}^{-1}$                   |
| Standard molar volume               |                                       | $22.414 \times 10^{-3} \text{ m}^3 \text{ mol}^{-1}$      |
| Molar gas constant                  | $R$                                   | $8.315 \text{ J mol}^{-1} \text{ K}^{-1}$                 |
| Fine structure constant             | $\frac{e^2}{4\pi\epsilon_0\hbar c} =$ | $\alpha$ $(137.04)^{-1}$                                  |
| Permittivity of free space          | $\epsilon_0$                          | $8.854 \times 10^{-12} \text{ F m}^{-1}$                  |
| Magnetic permeability of free space | $\mu_0$                               | $4\pi \times 10^{-7} \text{ H m}^{-1}$                    |
| Bohr magneton                       | $\mu_B$                               | $9.274 \times 10^{-24} \text{ A m}^2 \text{ or J T}^{-1}$ |
| Nuclear magneton                    | $\mu_N$                               | $5.051 \times 10^{-27} \text{ A m}^2 \text{ or J T}^{-1}$ |
| Neutron magnetic moment             | $\mu_n$                               | $-1.9130\mu_N$                                            |
| Proton magnetic moment              | $\mu_p$                               | $2.7928\mu_N$                                             |

**Useful equations****(1) Identities:**

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\sin(\theta + \phi) = \sin \theta \cos \phi + \cos \theta \sin \phi$$

$$\cos(\theta + \phi) = \cos \theta \cos \phi - \sin \theta \sin \phi$$

$$\cos^2 \theta + \sin^2 \theta = 1$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\sin 2\theta = 2 \cos \theta \sin \theta$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

**(2) Series expansions (valid for  $|x| < 1$ ):**

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2$$

$$+ \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

**(3) Integrals:**

$$\int_0^\infty x^n e^{-x} dx = n!$$

$$\int_{-\infty}^\infty e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}}$$

$$\int_{-\infty}^\infty x^{2n} e^{-\alpha^2 x^2} dx = \frac{(2n)!}{n! 2^{2n} \alpha^n} \sqrt{\frac{\pi}{\alpha}}$$

Indefinite (with  $a > 0$ ):

$$\begin{aligned}
 \int \frac{dx}{x^2 + a^2} &= \frac{1}{a} \tan^{-1} \frac{x}{a} \\
 \int \frac{dx}{x^2 - a^2} &= \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| \\
 \int \frac{dx}{\sqrt{x^2 + a^2}} &= \sinh^{-1} \frac{x}{a} \\
 \int \frac{dx}{\sqrt{x^2 - a^2}} &= \begin{cases} \cosh^{-1} \frac{x}{a} & \text{if } x > a \\ -\cosh^{-1} \frac{x}{a} & \text{if } x < -a \end{cases} \\
 \int \frac{dx}{\sqrt{a^2 - x^2}} &= \sin^{-1} \frac{x}{a}
 \end{aligned} \tag{G.1}$$

(4) **Vector operators:**

- grad acts on a scalar field to produce a vector field:

$$\text{grad } \psi = \nabla \psi = \left( \frac{\partial \psi}{\partial x}, \frac{\partial \psi}{\partial y}, \frac{\partial \psi}{\partial z} \right)$$

- div acts on a vector field to produce a scalar field:

$$\text{div } \mathbf{A} = \nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

- curl acts on a vector field to produce another vector field:

$$\text{curl } \mathbf{A} = \nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A_x & A_y & A_z \end{vmatrix}$$

where  $\psi(\mathbf{r})$  and  $\mathbf{A}(\mathbf{r})$  are any given scalar and vector field respectively.

(5) **Vector identities:**

$$\begin{aligned}
 \nabla \cdot (\nabla \psi) &= \nabla^2 \psi \\
 \nabla \times (\nabla \psi) &= 0 \\
 \nabla \cdot (\nabla \times \mathbf{A}) &= 0 \\
 \nabla \cdot (\psi \mathbf{A}) &= \mathbf{A} \cdot \nabla \psi + \psi \nabla \cdot \mathbf{A} \\
 \nabla \times (\psi \mathbf{A}) &= \psi \nabla \times \mathbf{A} - \mathbf{A} \times \nabla \psi \\
 \nabla \times (\nabla \times \mathbf{A}) &= \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \\
 \nabla \cdot (\mathbf{A} \times \mathbf{B}) &= \mathbf{B} \cdot \nabla \times \mathbf{A} - \mathbf{A} \cdot \nabla \times \mathbf{B} \\
 \nabla(\mathbf{A} \cdot \mathbf{B}) &= (\mathbf{A} \cdot \nabla) \mathbf{B} + (\mathbf{B} \cdot \nabla) \mathbf{A} \\
 &\quad + \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) \\
 \nabla \times (\mathbf{A} \times \mathbf{B}) &= (\mathbf{B} \cdot \nabla) \mathbf{A} - (\mathbf{A} \cdot \nabla) \mathbf{B} \\
 &\quad + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A})
 \end{aligned}$$

These identities can be easily proved by application of the alternating tensor and use of the summation convention. The alternating tensor  $\epsilon_{ijk}$  is defined according to:

$$\epsilon_{ijk} = \begin{cases} 1 & \text{if } ijk \text{ is an even permutation of } 123 \\ -1 & \text{if } ijk \text{ is an odd permutation of } 123 \\ 0 & \text{if any two of } i, j \text{ or } k \text{ are equal} \end{cases}$$

so that the vector product can be written

$$(\mathbf{A} \times \mathbf{B})_i = \epsilon_{ijk} A_j B_k.$$

The summation convention is used here, so that twice repeated indices are assumed summed. The scalar product is then

$$\mathbf{A} \cdot \mathbf{B} = A_i B_i.$$

Use can be made of the identity

$$\epsilon_{ijk} \epsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}$$

where  $\delta_{ij}$  is the Kronecker delta given by

$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

The vector triple product is given by

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C}) \mathbf{B} - (\mathbf{A} \cdot \mathbf{B}) \mathbf{C}.$$

(6) **Cylindrical coordinates:**

$$\nabla^2 \psi = \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \psi}{\partial \phi^2} + \frac{\partial^2 \psi}{\partial z^2}$$

$$\nabla \psi = \left( \frac{\partial \psi}{\partial r}, \frac{1}{r} \frac{\partial \psi}{\partial \phi}, \frac{\partial \psi}{\partial z} \right)$$

(7) **Spherical polar coordinates:**

$$\nabla^2 \psi = \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial \psi}{\partial \theta} \right)$$

$$+ \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2}$$

$$\nabla \psi = \left( \frac{\partial \psi}{\partial r}, \frac{1}{r} \frac{\partial \psi}{\partial \theta}, \frac{1}{r \sin \theta} \frac{\partial \psi}{\partial \phi} \right)$$

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